# Taxing Multinational Digital Platforms: Evidence from France's Digital Services Tax

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**Abstract**

Digital services taxes are levies on the revenues of large multinational digital platforms. Although digital taxes discourage platform investment and digital service provision, the accompanying reduced advertising can make websites more attractive to end-users. This paper examines the impact of France's 3% digital services tax, using monthly traffic data for 55 website families owned by 20 liable corporate groups. Effects of the tax vary by business model: web traffic declines by 11.1 log percentage points for pure intermediation websites and increases by 16.9 log percentage points for pure advertising websites. Traffic on nonliable sites that are more likely to be accessed via digital advertising declines by 8.2 log percentage points. These responses illustrate the potential for digital taxation to generate unintended consequences.

## 1 Introduction

Digital services taxes are levies on the gross local advertising and intermediation revenues of large multinational digital platforms. By distorting incentives to supply digital services, they lead platforms to adjust their investments into the diverse and idiosyncratic inputs into digital platform production, such as content production, data collection, and software engineering.

These adjustments are often difficult to observe directly. Public financial statements and investor reports only selectively reveal how platforms allocate expenditure across services and geographic markets. This

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is because global nature of platform operations and the nonrivalry of digital inputs severs the tie between the location of user activity and the jurisdictions where accounting items are recorded. Moreover, the platforms subject to DSTs often operate as "walled gardens": they control access to their services and the data generated by users, limiting what outside researchers can observe. These restrictions, together with stark differences in business models and pricing strategies, complicate the search for a common yardstick with which to compare platforms' responses to taxation.

In this paper, web traffic provides the necessary common measure that allows comparisons across countries, platforms, and over time. The key insight is that a user's decision to visit a platform's website reflects the private value of the information and services provided on that site. This value, in turn, depends on the aforementioned inputs into digital production. As a result, changes in traffic following a tax change provide useful evidence about the platform's behavioral response.

This paper starts by proposing a theoretical framework that ties unobservable platform choices to user demand. This is an important first step, because digital services taxes often target two tax bases that interact with user value in fundamentally different ways. While a turnover tax tends to depress the return to taxed activities and generally results in less, for example, digital advertising and e-commerce intermediation, the impacts of these changes on user traffic push in opposite directions. For an advertising-financed platform, user-facing expenditure attracts attention, while advertising monetizes that attention but discourages visits. A tax on advertising revenue reduces the return to providing content and services, but it can also induce the platform to reduce advertising intensity. The resulting traffic response depends on which adjustment dominates. For a platform that is primarily in the business of intermediating transactions, higher fees instead raise buyer prices and reduce seller supply, making the platform less attractive to users through both channels.

To test these theoretical predictions, this paper studies the evolution of traffic around the December 2018 announcement of France's 3 percent digital services tax on targeted advertising and digital intermediation. A heterogeneous difference-in-differences model is estimated for 55 popular websites owned by liable corporations, using synthetic controls constructed from nonliable sites without documented advertising exposure. Exposure to advertising is measured by computing, for each liable firm, the share of in-scope taxable revenues attributed in segment reports to digital advertising. The estimates indicate that a one-standard-deviation increase in this share—36.4 percentage points—is associated with a 12.38 log percentage point increase in traffic originating from France (s.e. 4.92). In these regressions, the effect of the tax on the average platform is $-2.81$ log percentage points (s.e. 6.99).

To put this estimate into context, Empirical Bayes is used to combine each website's observed-minus-synthetic traffic change with its prediction from the difference-in-differences model. This allows individual websites to depart from the fitted relationship while giving greater weight to more precise website-level estimates. This procedure indicates an increase of 16.9 log percentage points (s.e. 7.8) for website families whose parents' measured in-scope revenue comes entirely from advertising and a decline of 11.1 log percentage points (s.e. 8.2) for those whose parents' measured in-scope revenue comes entirely from intermediation.

This paper also studies the effect of digital service taxes on websites not directly liable for the digital services tax. Websites with observed advertising-related connections to liable platforms see declines of 8.24 log percentage points (s.e. 4.96) relative to sites without such observed connections. This decline is consistent with reduced advertising-induced traffic.

Seen through the lens of the conceptual framework sketched out above (and detailed in Section 3), the increase in traffic to liable advertising platforms, and the accompanying decline in traffic to likely advertisers, suggests that the digital services tax reduced not only the number of ads shown on liable digital platforms but also the number of ads shown per page load on these sites. Combining the estimated 8.24 log percentage point decline in total advertising with the 16.9 log percentage point increase in platform traffic implies a decline of approximately 25 log percentage points in ads shown per page load. Because cost minimization implies a unit elasticity of traffic with respect to advertising intensity, the model-implied residual contribution of reduced content and service provision to the traffic growth of advertising platforms is $-8.24$ log percentage points in the year after the DST was announced.

On the other hand, the decline in traffic to e-commerce intermediation platforms suggests that the French digital services tax made offerings on these platforms either more expensive or less varied, or both. The theoretical model connects the traffic estimates on intermediation platforms to existing evidence on consumer prices. Langenmayr and Muddasani (2025) estimate a French consumer log-price increase of 2.2 log percentage points, for products sold by third-party sellers using Fulfillment by Amazon following the DST. Combining this price response with this paper's estimated 4.50 log percentage point traffic decline for Amazon gives an upper bound of 2.05 on the elasticity of traffic with respect to tax-induced price changes. Studies of price sensitivity in e-commerce contexts suggest an absolute price elasticity of approximately 1.5. Assuming that price-related traffic declines occur in proportion to this demand elasticity, the model suggests that the residual one quarter of the traffic decline is attributable to reductions in product variety or seller participation.

The analysis presented here has important implications for proposals to use advertising taxes to counteract the harmful effects of social media (Romer, 2019; Acemoglu and Johnson, 2024).[1] The theoretical framework and empirical evidence presented herein show that by reducing advertising, digital services taxes, particularly those on advertising revenues, can make a platform more attractive to users, potentially increasing participation. The lesson is that taxes on digital advertising are a potentially counterproductive substitute for levies that directly target harmful use, as prescribed by the targeting principle (Sandmo, 1975; Kopczuk, 2003). This tradeoff illustrates the logic of the second best: when harmful platform use cannot be targeted directly, a tax that reduces advertising need not improve welfare because it also changes incentives for participation (Lipsey and Lancaster, 1956). Normative analyses of targeted advertising taxes should therefore be careful to distinguish the tax's potential corrective effect on the advertising market from its potential to

[1] These proposals, in turn, are informed by research documenting adverse effects of social media on mental health and well-being (Allcott et al., 2020; Braghieri et al., 2022), as well as evidence of habit formation and self-control problems that can lead users to consume more social media than they would otherwise prefer (Hoong, 2021; Allcott et al., 2022).

exacerbate overconsumption of digital services.

The policy implications extend to the debate about origin versus destination-based taxation of multinational digital business. This is because, as emphasized by Hines (2023), the multinational nature of location-based digital taxation, and in particular the public good nature of digital platform inputs, imply that taxing jurisdictions bear only a fraction of the cost associated with digital taxation. Such cross-border spillovers can create incentives for inefficiently high unilateral digital tax rates. This incentive is strongest when taxation imposes little domestic cost; the evidence in this paper shows that the French DST, in fact, generates substantial domestic costs, placing a natural limit on France's incentive to tax goods with global inputs.

The results of this study connect to research on indirect taxation of online commerce and platform-mediated transactions. Agrawal and Fox (2017, 2021) examine the collection and geographic assignment of consumption taxes in the digital economy, while studies of Airbnb and ride-hailing document how transaction taxes affect prices and market activity. Bibler et al. (2021) find that Airbnb tax enforcement raises prices paid by renters, and Tarduno (2025) finds substantial pass-through of ridesharing taxes to passengers. These studies establish that taxation affects the terms on which users transact through digital platforms. This study complements these findings and shows that the effects of digital taxation extend to a broader sample of liable websites.

This paper relates to the broader literature on taxation in multi-sided markets, surveyed by Kind and Schjelderup (2025), and the literature on multi-sided platforms as a whole (Rysman, 2009; Rochet and Tirole, 2003, 2006). Although this study examines platforms that serve multiple groups of users, an important intellectual contribution of the theoretical work is that none of the results rely on the distinctive pricing mechanisms emphasized in platform economics. In this framework, digital service provision is a cost of doing business, while the incidence of commissions and advertising costs reflects the balance of supply and demand in the market for the final good. The prediction that an advertising tax can increase traffic arises from the externality imposed on users who are shown advertisements they do not want to see. A tax that induces less advertising can reduce this externality and attract more visits, even as it lowers the return to providing services. Although this model-disciplined approach to interpreting the empirical estimates relies on structural assumptions about platform behavior, the central empirical argument is broader: changes in traffic provide evidence that platforms change their operations in response to taxation.

This paper proceeds as follows. Section 2 reviews the institutional context by providing a description of France's DST along with an exposition of liable platforms and their connection to nonliable websites. Section 3 outlines a theoretical model that connects web traffic to unobservable platform inputs via user preferences. Section 4 describes the data and empirical specifications and summarizes the results. Section 5 discusses the results in the context of the models in Section 3 and existing empirical research on DSTs. Section 6 concludes.

## 2 France's Digital Services Tax

France announced its digital services tax in December 2018 amid ongoing international negotiations within the OECD/G20 Inclusive Framework on Base Erosion and Profit Shifting (BEPS) over whether and how to expand countries' rights to tax multinational firms' profits derived from local activity (OECD, 2018). Existing rules generally tied taxing rights to firms' physical presence and allocated profits according to the location of their functions, assets, and risks. This allowed digital platforms to earn substantial revenues from local users while reporting little or no taxable profit in the countries where these users are located.[2]

The French DST was therefore intended to address these perceived shortcomings by taxing revenues associated with digital activity in France, pending coordinated reform of international corporate taxation that would resolve the issue of allocating profits.

The distinction between the intended object of taxation—profits attributable to economic activity in France—and the statutory tax base—gross revenues allocated to France—is consequential. Digital platforms often rely on substantial investments in intangible assets, such as software, algorithms, and digital interfaces, that can serve users across multiple jurisdictions at relatively low incremental cost. Because the costs of developing these assets are shared across markets, attributing a portion of worldwide profits to any particular jurisdiction requires potentially contentious allocation rules. By contrast, advertising placements, transactions, and other revenue-generating activities can more readily be assigned to the locations of users.

The choice to tax revenues rather than profits has important implications for production incentives. In a static context, a proportional tax on economic profits, with full deduction of economic costs, leaves a firm's optimal production decisions unchanged because it scales revenues and costs equally. A gross-revenue tax instead reduces the marginal return to production without reducing marginal costs, discouraging otherwise profitable activity.

There are good reasons to believe that this problem is less acute in the context of digital platforms, where the marginal cost of serving an additional user is commonly assumed to be negligible. In a conventional model of ad valorem revenue taxation, zero marginal production costs imply that the tax leaves a firm's revenue-maximizing choices unchanged.

The issue is that the production of digital services is not necessarily a zero marginal cost endeavor. In the context of social media platforms, even if serving an additional unit of content or advertising costs almost nothing, producing the content that attracts users in the first place is far from free. This is the force of Shapiro and Varian (1999)'s observation that information goods are costly to produce but cheap to reproduce. Moreover, sustaining user engagement requires ongoing investment in content, recommendations, and service quality rather than a single fixed expenditure. The economically relevant cost is therefore not simply the marginal cost of serving another visit, but the recurring cost of producing and maintaining the services that generate visits.

[2]Particular attention has been paid to the use of transfer pricing strategies to minimize corporate tax liabilities in high tax jurisdictions. See (Dharmapala, 2014).

The international reach of these investments may nevertheless attenuate this effect. Improvements to software, recommendations, or content can benefit users across multiple countries, so revenues from France represent only part of their return. To the extent that investment decisions are sustained by returns from other markets, a tax on French revenues may have a limited effect on these shared inputs.

In the context of e-commerce intermediation platforms, the services underlying digital production extend beyond displaying listings and matching buyers with sellers, which again are the activities thought to be zero marginal cost. Platforms process payments, manage fraud and risk, and provide support when transactions generate cancellations, refunds, or disputes. Platforms intermediating sales of physical goods may also provide warehousing, order fulfillment, shipping, and returns processing.

What this discussion is meant to emphasize is that the assumption of zero marginal costs is far from assured. Because the marginal conditions that determine whether a distortion is expected are fundamentally unobservable, the extent to which digital services taxes distort digital production is an empirical question that this paper seeks to address.

## 2.1 Policy Details

The French DST imposes a 3% tax on gross revenues attributable to France from specified digital activities. For 2019, the tax applied to firms or corporate groups whose revenues from taxable services in 2018 exceeded both €750 million worldwide and €25 million in France. These thresholds are assessed using taxable-service revenues aggregated across entities within a corporate group.

The tax distinguishes two principal categories of digital services. Targeted advertising encompasses the placement of advertisements based on data collected from users of digital interfaces, as well as related advertising services and the sale of user data for advertising purposes. Digital intermediation encompasses services that enable users to interact or transact through a digital interface, with taxable revenues consisting of fees and other payments received in exchange for providing these services. Because the DST is levied on gross revenues, firms cannot deduct the costs incurred in providing taxable services. The tax base may also include payments for complementary services that are not economically independent of access to or use of a digital interface, such as fulfillment services bundled with marketplace access.

The DST attributes revenues to France using a service-specific measure of French user activity. For each taxable service, the French tax base is calculated by multiplying worldwide taxable revenues by a French presence coefficient, which measures the share of relevant activity attributable to France. For targeted advertising, this coefficient equals the share of advertisements targeted toward users located in France, rather than the proportion of advertising expenditures attributable to France. For online marketplaces, it equals the share of transactions involving at least one user located in France. For other intermediation services, it is based on the share of active users whose accounts were opened from France.

Accordingly, a firm's DST liability can be expressed as

$$T = \tau \sum_{s \in \mathcal{S}} R_s \alpha_s, \tag{1}$$

where $\tau = 0.03$, $R_s$ denotes worldwide taxable revenue from service $s$, and $\alpha_s$ is its French presence coefficient.

This equation underscores the underappreciated fact that the French digital service tax is not, in fact, strictly a tax on local revenues, but rather a tax on the French portion of global revenues, where the presence share is defined as a function of the firm's interactions with users rather than the location of revenues. In the context of the French government's justification for the tax, this may appear sensible, but it introduces additional distortions beyond those created by the decision to tax gross revenues rather than profits. To see this, consider the effective marginal tax on an advertisement sold in France.

$$t_F^{\text{eff}} = \tau \left[ \alpha_A + (1 - \alpha_A) \frac{\bar{p}}{p_F} \right], \tag{2}$$

Here, $\alpha_A$ is the French share of worldwide advertising impressions, $\bar{p}$ is average worldwide revenue per impression, and $p_F$ is the revenue from an additional impression shown in France. The effective marginal rate exceeds the statutory 3 percent when $p_F < \bar{p}$, falls below it when $p_F > \bar{p}$, and equals it when the two coincide. The difference arises because an additional French impression increases both worldwide advertising revenue and the share allocated to France. Although one impression changes the allocation share only slightly, that change applies to the entire worldwide revenue base.

Equation 1 also creates an implicit distortion on ads and intermediation that take place outside of France, because an additional foreign impression can reduce French tax liability when its revenue is below the worldwide average. Again, consider the effective marginal tax on an advertisement shown outside of France.

$$t_R^{\text{eff}} = \tau \alpha_A \left( 1 - \frac{\bar{p}}{p_R} \right), \tag{3}$$

where all notation is as before except that $p_R$ represents the return on an ad shown outside of France.

Estimating these effective marginal tax rates is difficult because neither service-specific presence ratios nor relative advertising returns are directly observed. Table 1 presents suggestive calibrations for Facebook, Instagram, and LinkedIn, combining Statista's modeled French shares of worldwide users in 2018 with published cost per mille (CPM) benchmarks. The implied marginal rates on advertisements shown in France range from 3.40 to 8.84 percent, substantially higher than the statutory rate of 3 percent. Rates on advertisements shown in Germany and the United States are much smaller, with negative rates for Germany and generally positive rates for the United States.

Table 1: Calibrated marginal French DST rates using Statista usage shares and CPM benchmarks

| | | | | Marginal rate (%) | | |
|---|---|---|---|---|---|---|
| Product | CPM period | $\alpha_F$ (%) | $q_F$ | France | Germany | US |
| Facebook | 2015 Q4 | 1.47 | 0.336 | 8.84 | -0.020 | +0.011 |
| Instagram | 2015 Q4 | 1.75 | 0.881 | 3.40 | -0.006 | -0.017 |
| Facebook | 2016 Q2 | 1.47 | 0.499 | 5.96 | -0.012 | +0.006 |
| Facebook, app installs | 2016 Q2 | 1.47 | 0.482 | 6.17 | -0.011 | +0.009 |
| LinkedIn, single image | 2025–26 | 2.78 | 0.568 | 5.22 | -0.021 | +0.034 |

*Notes:* Rates are percentages of marginal advertising revenue; negative entries indicate implicit subsidies. $\alpha_F$ is the platform-specific ratio of modeled French to worldwide users in the 2018 Key Market Indicators estimates of Statista (2026a,b,c): Facebook 1.475%, Instagram 1.750%, and LinkedIn 2.783%. These audience shares proxy advertising-impression shares; they are not the parent-level traffic shares from the estimation sample. Statista's geographic modeling also uses traffic information, and the activity-frequency, reference-date, and platform-scope definitions remain imperfectly documented. $q_F$ is French CPM divided by the source's multinational CPM benchmark. CPM denotes cost per thousand impressions. Sources for prices are Salesforce Marketing Cloud (2016a,b) for the 2015–16 benchmarks and Kiin Labs (2026) for the 2025–26 LinkedIn benchmarks. Calculations substitute source benchmarks for worldwide average prices and average prices for marginal prices. Periods in the table describe price evidence; all usage shares refer to 2018. These are conditional calibrations rather than measured 2018 tax rates; no sampling confidence intervals are available.

## 2.2 Timeline

The French DST emerged from earlier European efforts to reform the taxation of digital services. In March 2018, the European Commission proposed a common 3% tax on revenues from certain digital activities. However, the proposal failed to secure the unanimous support required for adoption. Following unsuccessful negotiations in December 2018, France announced that it would proceed unilaterally. On December 17, 2018, Finance Minister Bruno Le Maire announced that France would introduce a tax covering digital advertising, online marketplaces, and the commercialization of user data, with revenues taxable beginning January 1, 2019. The government introduced legislation in March 2019, and the DST was enacted in July 2019 with retroactive application to revenues earned from the beginning of the year.

Although the tax applied to 2019 revenues, collection and enforcement were subsequently affected by international negotiations. The first installment was payable in November 2019. In January 2020, France and the United States reached an agreement to defer certain subsequent payments while negotiations over international tax reform continued. These deferrals did not eliminate the underlying tax liability, and the deferred installments became payable in December 2020. The distinction between liability and collection is important because firms' incentives to adjust their behavior depend on expected tax obligations, not necessarily the dates on which payments are made.

The DST was also introduced as an interim measure pending a multilateral agreement on international corporate taxation. In October 2021, France joined an agreement establishing transitional arrangements for existing digital services taxes in anticipation of the implementation of the OECD's Pillar One framework. The agreement contemplated the eventual withdrawal of unilateral measures and conditional credits against future tax liabilities. These developments could affect firms' expectations about the duration of the DST,

even while the tax remained in effect.

Figure 1 summarizes these developments, distinguishing the announcement and enactment of the tax, the timing of collections, and subsequent international agreements.

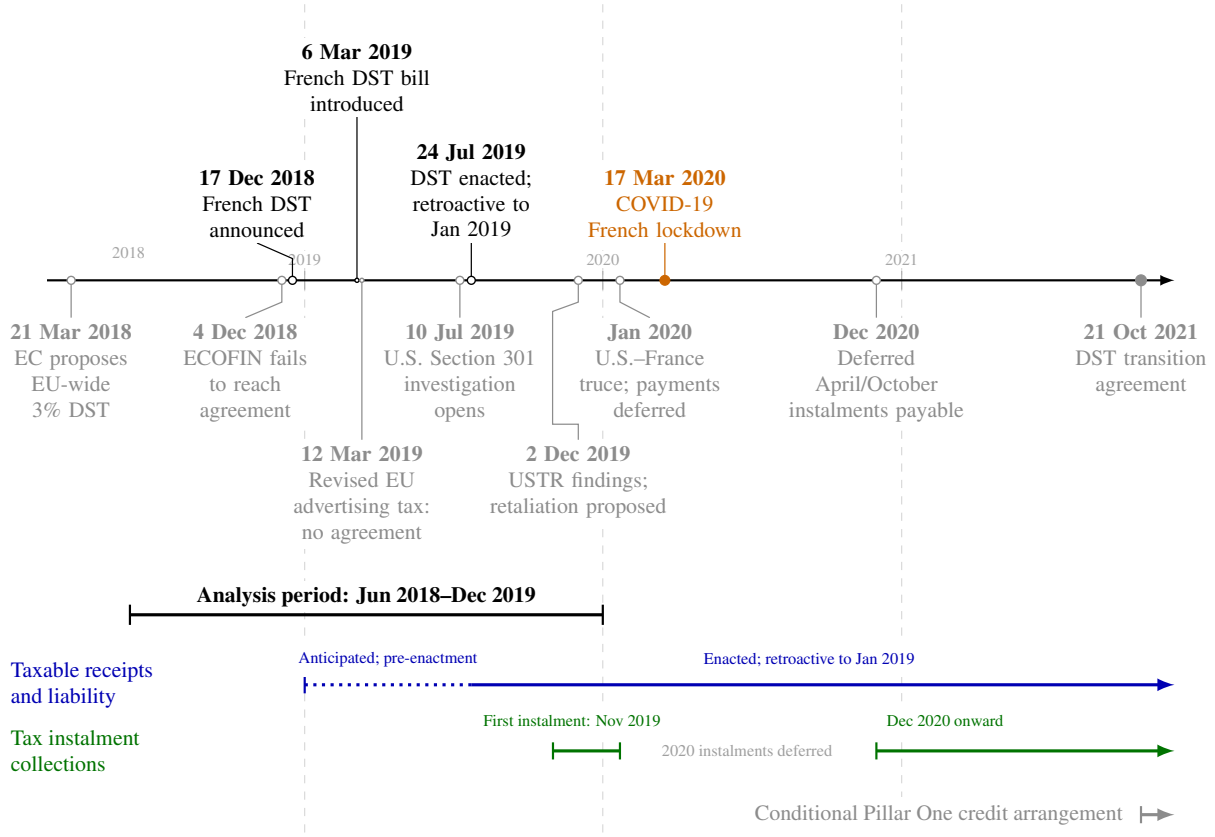



Figure 1: Timeline of the French digital services tax.

*Notes:* Event markers and period boundaries use proportional calendar-day spacing; labels are staggered for readability. Month-only payment events use the first of the month. The bracket covers June 1, 2018 through December 31, 2019 (right boundary: January 1, 2020). The dotted segment shows receipts subsequently covered by the law and anticipated liability before enactment; the solid segment shows enacted liability. The instalment-collection bars summarize instalment-collection periods, not continuous payments or all settlements. The first instalment was payable in November 2019. The optional deferral covered the April and October 2020 instalments, payable in December 2020; it did not cancel liability or imply that every other payment was suspended. The October 21, 2021 agreement provided conditional credits against future Pillar One Amount A liability; the specified interim credit period began January 1, 2022. Sources: European Commission (2018) and Council (2019); Assemblee nationale (2018); French Law No. 2019-759 (2019); DGFiP (2019; 2021); USTR (2021); and U.S. Treasury (2021).

## 2.3 Affected firms and their responses

The revenue thresholds used to determine corporate group-level liability make it so that only a handful of companies are thought to be subject to the French DST. This section identifies a group of 35 corporate groups considered likely liable.

The starting point is a list of 27 corporate groups listed by the U.S. Trade Representative's report on France's DST (United States Trade Representative, 2019). Reasonable minds may disagree on the accuracy of this list, given the report's conclusion that the French DST was designed to unfairly target American corporations. Concerns in this regard can be separated into two forms of misclassification: false positives and false negatives.

To screen for false positives in the USTR classification, fiscal-year 2018 consolidated revenue data were

collected from publicly available corporate accounts and complemented with French-entity turnover records from Orbis.[3] Columns 1 and 2 of Table 2 report these measures. All 27 groups have total consolidated revenues above the global threshold. Among the 24 groups with positive recorded French-entity turnover, 17 exceed the local threshold.

Columns 3 and 4 improve on this rough test by showing revenues from advertising and intermediation, the two tax bases, taken from segment reports. This provides a deeper view into the consolidated revenue shares by tax base. 23 of the 25 groups with available data pass the in-scope revenue thresholds. The two that do not are Match Group and Zalando. For Match Group, the ad plus intermediation segment reports omit an unusual third category of subscription revenues that is likely in the liable tax base. Taking those into account raises its global revenues past the threshold. Zalando, on the other hand, is more likely to fail the global in-scope revenue tests, because its large size reflects the fact that most transactions that happen on their platform are first party.

The issue of false negatives in the USTR classification is more challenging because it requires assessing a large and not well defined set of candidate firms. In order to make this problem tractable, a deliberately over-inclusive screening list of 130 groups was first compiled from public sources, with assistance from a large language model. Then, firms sharing at least one in-scope industry code with a firm on this list form the candidate universe. The consolidated revenue thresholds and, when available, the French-specific revenue thresholds determine the selected subset. A subsequent review of these candidates' business activities and available evidence on in-scope revenues identified eight additional corporate groups that are likely liable for the French DST.[4]

In addition to the search for liable groups detailed above, the text of publicly available corporate filings and investor reports was searched for mentions of the French DST. This search provided additional evidence of DST liabilities for four firms in this list: Booking, ANGI, TripAdvisor, and Match. See Appendix B for more details on this exercise.

Firms' announced responses show how the tax can change the terms offered to other platform participants. In August 2019, Amazon stated that certain seller fees for sales on Amazon France would increase by 3 percent from October 1, 2019 (Amazon, 2019). This was a proportional increase in fees, rather than a three-percentage-point increase in the commission rate. Apple's September 2020 notice stated that developer proceeds would be adjusted to account for France's 3 percent DST while posted App Store prices would remain unchanged, shifting the charge to developers through lower proceeds (Apple, 2020).

Advertising platforms also introduced explicit surcharges. Google added a 2 percent regulatory operating cost to advertisements served in France from May 1, 2021, describing it as covering part of the cost of complying with the French DST (Google, 2021). Meta later announced a 3 percent location fee for advertisements delivered to users in France, effective July 1, 2026 (Adegbola, 2026). These announcements

[3] Appendix A describes the data collection and its limitations.

[4] See Appendix A for details on the screening list and candidate-universe construction, including the USTR and Deloitte materials (United States Trade Representative, 2019; Pellefigue, 2019).

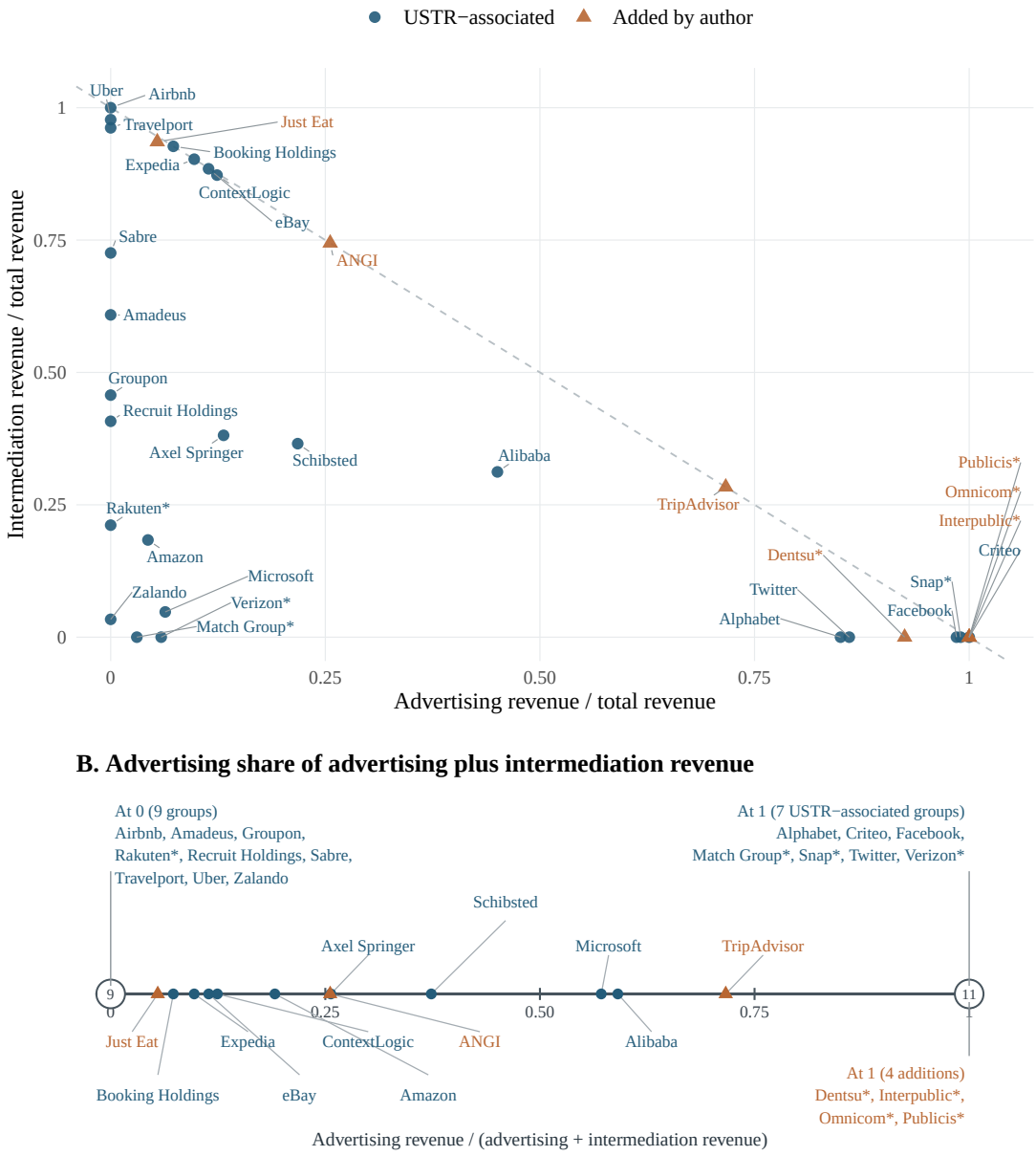
document changes in contractual charges, while their ultimate incidence depends on subsequent responses by sellers, developers, advertisers, and users. Amazon's announced change falls within the paper's 2019 post-announcement window; the Apple, Google, and Meta changes illustrate subsequent adjustments.

The other thing to observe in Table 2 is the revenue share columns 5 through 7. These data are presented visually in Figure 2 for readability. Panel A of Figure 2 shows that although many firms operate almost exclusively in the in-scope markets, a substantial share of them have significant revenues in other businesses. Most often, this third line of business is direct retail sales of software, hardware, and other goods. There are also notable cases where the parent is a big player in the cloud computing business, and one case where the parent company's revenues primarily reflect telecom income. Panel B shows how the firms map onto the two taxable revenue categories as a share of each firm's DST-taxable income. The data show that about one-third of the firms only participate in advertising, one-third only participate in intermediation, and the final third fall somewhere in between.

The next task is to map corporate groups to websites owned by them before treatment. Table 3 lists these corporate groups and the websites they owned before treatment. It also reports November 2018 French traffic ranks and lists the other consumer websites linked to each group that are not in the analysis sample. [5]

Table 4 reports the treated websites' share of pre-treatment French and global traffic within the top 100, 1,000, and 2,000 website families. This information shows that despite the small number of corporate groups and websites considered liable for the French DST, they constitute a very large share of traffic to the top websites in France and the world. Finally, Panel B and Figure 3 summarize the relative "Frenchness" of websites in the sample, ranking them by their estimated French share of global traffic. These measures capture differences in geographic orientation within the sample rather than the French presence ratios used to calculate tax liability.

[5] Appendix A documents the evidence on likely liability and the links from firms to observed websites.




FY2018. Equal−size corporate−group markers; 32 groups with available shares.
Apple, Randstad and Cheil are omitted. * indicates a proxy or classification limitation; see figure notes.

Figure 2: Corporate revenue composition, FY2018

*Notes:* Panel A plots FY2018 advertising and intermediation revenues as shares of FY2018 consolidated revenue; the dashed line indicates that the two sum to total revenue. Panel B places advertising's share of their combined revenue on a number line; endpoint circles report group counts. The omitted category primarily includes first party sales and cloud computing revenues. Labels are offset for readability. Blue circles denote USTR-associated groups; orange triangles denote the author's additions. The figure preserves the classifications in Table 2; Appendix A.5 documents the company filings, Orbis data (Bureau van Dijk, a Moody's Analytics company, 2025) and revenue classifications. Asterisks flag limitations: Match's dating subscriptions are coded outside intermediation; Rakuten's advertising is not separately allocated; Verizon uses Oath revenue; Snap uses its rounded 99% advertising share; agency revenues include traditional advertising. Apple, Randstad, and Cheil are omitted because their splits are missing.

Table 2: Corporate revenues and business composition, FY2018

| Corporate group | Revenue (EUR millions) | | | | Revenue shares (%) | | | |
|---|---|---|---|---|---|---|---|---|
| | Consolidated total | French entities | Ads | Inter-mediation | Ads/ total | Interm./ total | Ads/ (ads+interm.) | French/ total |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Panel A. USTR-associated groups (27)* | | | | | | | | |
| Airbnb | 3,099.6 | 14.3 | 0.0 | 3,099.6 | 0.0 | 100.0 | 0.0 | 0.46 |
| Alibaba | 48,385.1 | 5.4 | 21,796.6 | 15,101.6 | 45.0 | 31.2 | 59.1 | 0.01 |
| Alphabet | 116,125.7 | 412.2 | 98,725.4 | 0.0 | 85.0 | 0.0 | 100.0 | 0.35 |
| Amadeus | 4,935.7 | 1,713.8 | 0.0 | 3,004.3 | 0.0 | 60.9 | 0.0 | 34.72 |
| Amazon | 197,663.8 | 735.1 | 8,579.2 | 36,280.0 | 4.3 | 18.4 | 19.1 | 0.37 |
| Apple | 225,424.9 | 930.8 | – | – | – | – | – | 0.41 |
| Axel Springer | 3,180.7 | 110.3 | 418.3 | 1,212.5 | 13.2 | 38.1 | 25.6 | 3.47 |
| Booking Holdings | 12,329.9 | 35.7 | 899.7 | 11,430.2 | 7.3 | 92.7 | 7.3 | 0.29 |
| ContextLogic | 1,466.6 | – | 181.6 | 1,279.9 | 12.4 | 87.3 | 12.4 | – |
| Criteo | 1,952.4 | 875.7 | 1,952.4 | 0.0 | 100.0 | 0.0 | 100.0 | 44.85 |
| eBay | 9,120.7 | 6.6 | 1,039.7 | 8,068.3 | 11.4 | 88.5 | 11.4 | 0.07 |
| Expedia | 9,525.6 | 158.2 | 926.8 | 8,598.7 | 9.7 | 90.3 | 9.7 | 1.66 |
| Facebook | 47,392.7 | 390.4 | 46,692.5 | 0.0 | 98.5 | 0.0 | 100.0 | 0.82 |
| Groupon | 2,237.9 | 41.9 | 0.0 | 1,023.2 | 0.0 | 45.7 | 0.0 | 1.87 |
| Match Group | 1,468.2 | 27.9 | 44.7 | 0.0 | 3.0 | 0.0 | 100.0 | 1.90 |
| Microsoft | 93,668.5 | 1,439.7 | 5,951.5 | 4,463.6 | 6.4 | 4.8 | 57.1 | 1.54 |
| Rakuten | 8,474.1 | – | 0.0[c] | 1,792.6 | 0.0 | 21.2 | 0.0 | – |
| Randstad | 23,812.0 | 3,938.8 | – | – | – | – | – | 16.54 |
| Recruit Holdings | 17,777.4 | 741.5 | 0.0 | 7,248.7 | 0.0 | 40.8 | 0.0 | 4.17 |
| Sabre | 3,282.1 | 8.3 | 0.0 | 2,381.8 | 0.0 | 72.6 | 0.0 | 0.25 |
| Schibsted | 1,877.8 | 417.7 | 409.1 | 686.3 | 21.8 | 36.5 | 37.3 | 22.24 |
| Snap | 1,001.9 | 46.7 | ≈991.9 | 0.0 | ≈99.0 | 0.0 | 100.0 | 4.66 |
| Travelport | 2,165.2 | 11.4 | 0.0 | 2,082.9 | 0.0 | 96.2 | 0.0 | 0.53 |
| Twitter | 2,582.2 | 13.3 | 2,221.5 | 0.0 | 86.0 | 0.0 | 100.0 | 0.52 |
| Uber | 9,565.5 | 58.2 | 0.0 | 9,349.0 | 0.0 | 97.7 | 0.0 | 0.61 |
| Verizon | 111,070.5 | 4.7 | 6,535.4[a] | 0.0 | 5.9 | 0.0 | 100.0 | $< 0.01$ |
| Zalando | 5,387.9 | – | 0.0 | 181.4 | 0.0 | 3.4 | 0.0 | – |
| *Panel B. Groups added by the author (8)* | | | | | | | | |
| ANGI | 961.0 | – | 245.7 | 715.3 | 25.6 | 74.4 | 25.6 | – |
| Cheil | 2,716.6[b] | 54.7 | – | – | – | – | – | 2.01 |
| Dentsu | 7,835.8 | 312.6 | 7,246.6[a] | 0.0 | 92.5 | 0.0 | 100.0 | 3.99 |
| Interpublic | 8,245.1 | 191.5 | 8,245.1[a] | 0.0 | 100.0 | 0.0 | 100.0 | 2.32 |
| Just Eat | 879.4 | 43.0 | 47.7 | 822.7 | 5.4 | 93.6 | 5.5 | 4.89 |
| Omnicom | 12,977.6 | 493.3 | 12,977.6[a] | 0.0 | 100.0 | 0.0 | 100.0 | 3.80 |
| Publicis | 9,951.0 | 1,155.6 | 9,951.0[a] | 0.0 | 100.0 | 0.0 | 100.0 | 11.61 |
| TripAdvisor | 1,370.7 | – | 982.0 | 388.7 | 71.6 | 28.4 | 71.6 | – |

*Notes:* Columns (1), (3), and (4) are FY2018 consolidated revenues from company filings, in euros. Column (2) is turnover of the group's French entities from unconsolidated Orbis accounts; it is not sales to French users or the DST base. Columns (5)–(7) are $100 \times (3)/(1)$, $100 \times (4)/(1)$, and $100 \times (3)/[(3) + (4)]$; column (8) is $100 \times (2)/(1)$, calculated before rounding. Dashes indicate unavailable splits or accounts; zeros indicate no revenue allocated to the category. [a]Includes traditional advertising (agencies) or uses Oath revenue (Verizon). [b]From Orbis. [c]Not separately allocated. ≈ Approximated from a reported share. Sources: company filings, Orbis (Bureau van Dijk, a Moody's Analytics company, 2025), and United States Trade Representative (2019). Appendix A.5 documents the sources, revenue classification, exchange rates, and each adjusted entry.

Table 3: Corporate groups and their linked websites

| Corporate group | Websites in the analysis sample | Other linked consumer websites not in the analysis sample |
|---|---|---|
| Airbnb | Airbnb (102) | – |
| Alibaba | AliExpress (11), Alibaba (350), Taobao (768) | Lazada (1,200), Tmall (1,296), Youku (1,550), 1688 (1,745), Trendyol (n.a.) |
| Alphabet | Google (1), YouTube (2) | Googleusercontent (399), Android (500), Blogger (630), Kaggle (743), Tenor (n.a.), Waze (n.a.) |
| Amadeus | – | – |
| Amazon | Amazon (4), Twitch (37), IMDb (180), Gamepedia (212), Goodreads (441), CurseForge (470), Futhead (564), DPReview (718), HearthPwn (918) | Prime Video (227), Souq (1,282), Audible (1,347), Woot (2,157), Mechanical Turk (2,197), 6pm (n.a.), AbeBooks (n.a.), Book Depository (n.a.), Box Office Mojo (n.a.), Shopbop (n.a.), Zappos (n.a.), ZVAB (n.a.) |
| ANGI | – | HomeAdvisor (n.a.) |
| Apple | – | Apple (380), iCloud (494) |
| Axel Springer | La Centrale (79), SeLoger (84), Logic-Immo (264), Welt (990), Computer Bild (1,147), Bild (1,174) | Caradisiac (147), Idealo (1,039), Finanzen (1,164), Business Insider (1,213), Immoweb (1,382), Immowelt (1,392), Auto Bild (1,522), meinestadt (1,604), Transfermarkt (1,688), StepStone (1,704), kaufDA (1,951), Immonet (1,960), Travelbook (2,212), B.Z. (n.a.), Stylebook (n.a.), Totaljobs (n.a.) |
| Booking Holdings | Booking.com (101), Rentalcars.com (808) | Agoda (870), Kayak (1,535), Momondo (2,212), OpenTable (n.a.), Priceline (n.a.), Swoodoo (n.a.) |
| Cheil | – | – |
| ContextLogic | Wish (218) | – |
| Criteo | – | – |
| Dentsu | – | – |
| eBay | eBay (8), mobile.de (379), eBay Kleinanzeigen (432), 2ememain (505), Marktplaats (800), Kijiji (906) | Gumtree (925), 2dehands (1,249), DBA (1,627), Motor-Talk (1,695), Bilbasen (n.a.), brands4friends (n.a.), eBay Stores (n.a.), StubHub (n.a.) |
| Expedia | Hotels.com (515) | Expedia (1,327), Vrbo (2,034), FeWo-direkt (2,091), Abritel (n.a.), HomeAway (n.a.), Orbitz (n.a.), Travelocity (n.a.), Trivago (n.a.) |
| Facebook | Facebook (3), Instagram (53) | Messenger (196), WhatsApp (667) |
| Groupon | – | Groupon (1,439) |
| Interpublic | – | – |
| Just Eat | – | Just Eat (n.a.) |
| Match Group | Meetic (281), OkCupid (507), Plenty of Fish (538), Twoo (776) | Tinder (806), Match.com (2,171), DisonsDemain (n.a.), LoveScout24 (n.a.), OurTime (n.a.) |
| Microsoft | Live (18), MSN (24), LinkedIn (34), Bing (91), SlideShare (529) | GitHub (26), Microsoft (47), Office (85), SharePoint (161), Microsoft Online (265), Office 365 (371), Skype (497), Visual Studio (508), Xbox (568), Minecraft (678), Lynda.com (1,063), GroupMe (n.a.), Mixer (n.a.), Mojang (n.a.), Outlook (n.a.), Power BI (n.a.) |
| Omnicom | – | – |
| Publicis | – | – |
| Rakuten | Viki (548) | Rakuten (60), Infoseek (1,622), OverDrive (1,801), PriceMinister (2,099), Ebates (n.a.), Kobo (n.a.) |

*Continued on next page*

Table 3: Corporate groups and their linked websites (continued)

| Corporate group | Websites in the analysis sample | Other linked consumer websites not in the analysis sample |
|---|---|---|
| Randstad | – | Military.com (n.a.), Monster (n.a.) |
| Recruit Holdings | Indeed (78) | Glassdoor (1,566), Jalan (n.a.), SUUMO (n.a.) |
| Sabre | – | – |
| Schibsted | Leboncoin (9), Subito (766), Milanuncios (1,133) | Willhaben (1,306), InfoJobs (1,454), FINN (1,472), Blocket (1,583), Aftonbladet (1,606), OLX (1,756), VG (1,779), Shpock (1,824), Kufar (1,934), Tori (2,171), Aftenposten (n.a.), AVendreALouer (n.a.), Coches.net (n.a.), E24 (n.a.), Hasznaltauto (n.a.), Jofogas (n.a.), SvD (n.a.), Videdressing (n.a.) |
| Snap | – | – |
| Travelport | – | – |
| TripAdvisor | TripAdvisor (157) | Cruise Critic (n.a.), Viator (n.a.) |
| Twitter | Twitter (15) | Periscope (496) |
| Uber | Uber (665) | Uber Eats (n.a.) |
| Verizon | Yahoo (13), Huffington Post (138), Tumblr (143), AOL (318) | Oath (348), TechCrunch (1,065), Engadget (1,135), Verizon Wireless (1,658), Autoblog (n.a.), MapQuest (n.a.), Rivals (n.a.) |
| Zalando | Zalando (95) | Zalando Lounge (n.a.) |

*Notes:* The table lists the 185 consumer-facing website families linked to roster groups after Filters 1 and 2 of Appendix A.7. The middle column contains the 55 families in the analysis sample, belonging to 20 of the 35 groups. The last column contains the other 130, which are removed by Filters 3–5: no French traffic in April–November 2018, fewer than four French months in that period, or the activity review. Groups with a dash in both columns own no domain in the WTM universe (Amadeus, Travelport, and the five advertising agencies) or only non-consumer domains (Criteo, Sabre, and Snap). Ownership follows the December 2018 mapping described in Appendix A. Parentheses give November 2018 French traffic ranks among all 2,230 website families observed in WhoTracksMe that month (1 = highest traffic), after summing observed member-domain popularity. Families combine country-specific domains; ranks are not restricted to liable websites. "n.a." means no November 2018 French observation; a dash means no website in that column. Sources: WhoTracksMe (Karaj et al., 2018) and United States Trade Representative (2019); see Appendix A.

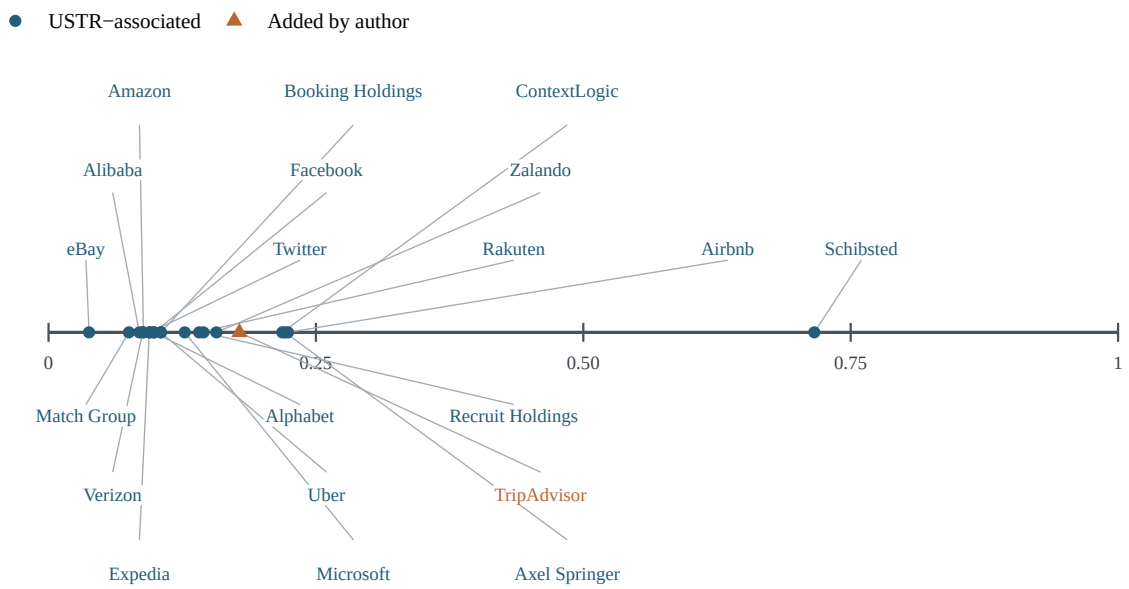



Figure 3: French shares of global traffic in WTM data

*Notes:* Each marker places one of the 20 treated corporate groups on a common number line from 0 to 1. Traffic covers the group's selected analysis-sample website families in September–November 2018. French popularity is calibrated monthly assuming that 90% of Amazon.fr's global traffic originates in France; each group share divides summed calibrated French popularity by summed global popularity across its sampled families and months. Shares describe sampled websites rather than all group properties. Blue circles identify USTR-associated groups; the orange triangle identifies the group added by the author. Labels are offset solely for readability. Missing observations are omitted. Source: author's calculations on WhoTracksMe data (Karaj et al., 2018).

Table 4: Pre-treatment traffic coverage and French shares of global traffic in the WTM data

*Panel A. Treated traffic within the top $N$ website families*

| Top $N$ families | France | | Global | |
|---|---|---|---|---|
| | Treated families | Traffic share (%) | Treated families | Traffic share (%) |
| 100 | 19 | 56.49 | 24 | 70.25 |
| 1,000 | 52 | 44.15 | 46 | 51.90 |
| 2,000 | 55 | 43.78 | 52 | 47.52 |

*Panel B. Estimated French share of global traffic (%)*

| Sample | $n$ | Mean | SD | Min. | P25 | Median | P75 | Max. |
|---|---|---|---|---|---|---|---|---|
| Website families | 55 | 16.28 | 26.00 | 0.23 | 4.50 | 8.81 | 14.28 | 96.77 |
| Corporate groups | 20 | 15.50 | 14.18 | 3.79 | 9.29 | 10.50 | 16.24 | 71.60 |

*Notes:* Panel A uses November 2018 WhoTracksMe popularity and the family definitions in Table 3. France (2,230 families) and the source's global series (2,382 families) are ranked independently; both use the same 55 treated families. Panel A divides treated popularity within each top $N$ by all popularity within that top $N$; cutoff ties break alphabetically. Panel B uses September–November 2018. In month $t$, French popularity is multiplied by $k_t = 0.9\, G_{\text{amazon.fr},t}/F_{\text{amazon.fr},t}$, assuming 90% of Amazon.fr's global traffic originates in France. Each share is $100 \sum_t k_t F_{it} / \sum_t G_{it}$. Group shares first sum traffic across the group's sampled families; they do not cover all group properties. Rows weight families or groups equally. SD is the sample standard deviation (percentage points in Panel B); quartiles use linear interpolation. Missing observations are omitted. Source: WhoTracksMe (Karaj et al., 2018) and the treatment roster documented in Appendix A.

## 2.4 Advertising Connections beyond Liable Platforms

The reach of liable platforms extends beyond the websites they own. Nonliable websites may purchase advertising from these platforms to attract visitors, supply advertising inventory through them, or do both. These connections provide channels through which the DST can affect websites outside its statutory scope.

Although historical advertising campaign records are unavailable, embedded third-party technologies reveal potential connections between websites and advertising providers. On an advertiser's website, JavaScript tags and tracking pixels can record visits and actions such as viewing a product or completing a purchase. Cookies and identifiers transmitted with these requests can allow advertising providers to associate these actions with earlier ad interactions or use them to select subsequent advertisements. For example, Facebook Pixel supports retargeting based on a visitor's interactions with an advertiser's website, while Bing Ads' Universal Event Tracking supports conversion measurement and remarketing. The database contains the broader Facebook and Bing Ads tracker identities, as well as Google AdServices and Criteo.[6]

Other technologies support the sale and delivery of advertising on third-party websites. The database identifies Google Syndication, Amazon Advertising, AppNexus, and Outbrain, among others. These

[6]Facebook's April 2018 description (Facebook, 2018) gives a retail retargeting example using Facebook Pixel. Microsoft's UET documentation (Microsoft, 2026) describes conversion tracking and remarketing; the latter is current product documentation, rather than a reconstruction of a particular website's November 2018 implementation.

services include publisher advertising networks, bidding infrastructure, and sponsored recommendations. Some providers serve both advertisers and publishers, so a provider's presence alone does not determine whether the host website buys advertising, sells advertising, or uses a related service.[7]

Not every third-party connection establishes advertising participation. A request for an image or another externally hosted resource can disclose a visitor's presence and, depending on the request, transmit an identifier. But general content-delivery services, standalone analytics, and tag-management containers also support activities unrelated to advertising. For the empirical measures, WTM's `advertising` category is used to identify advertising-related technologies, excluding other categories and the former custom narrow/broad overrides.

These technologies can support different sides of the advertising market: displaying or monetizing ads on the host website, or tracking conversions and enabling remarketing associated with ads shown elsewhere. The provider-level identities recorded in the data generally do not distinguish these uses. A connection to an advertising network may also reflect measurement or identifier matching, so its presence does not by itself establish that the host displays ads. Likewise, conversion-tracking infrastructure does not establish that the host purchased advertising. These observations are therefore interpreted as advertising-related technical connections without assigning each host to an advertiser or publisher role.

Table 5 reports November 2018 coverage among the paper's nonliable website families. Panel A lists the fifteen leading advertising-related technologies and their corporate owners, ranked by the share of global nonliable traffic accounted for by families with a published detection. The columns show the percentage of nonliable families with a detection and their shares of global and French-origin nonliable traffic. Panel B reports combined coverage across all advertising-related technologies, separating liable-owned from nonliable-owned providers and unresolved ownership. A detection on any member domain identifies a connection for its family.

[7] Amazon's January 2018 announcement (Amazon, 2018) describes its publisher bidding service and its availability in France. Criteo's 2018 annual report (Criteo, 2019, pp. 15–16) describes its advertiser and publisher services.

Table 5: Advertising-related technologies among nonliable websites, November 2018

| | | Sites (%) | | Traffic to identified sites (%) | |
|---|---|---|---|---|---|
| Technology | Corporate owner | Global | France | Global | France |
| *A. Top fifteen advertising-related technologies* | | | | | |
| Google | Alphabet* | 84.0 | 52.4 | 88.9 | 83.9 |
| DoubleClick | Alphabet* | 73.8 | 43.7 | 80.1 | 79.3 |
| Facebook | Facebook* | 59.7 | 42.8 | 62.7 | 64.6 |
| Google AdServices | Alphabet* | 31.0 | 18.7 | 37.8 | 41.4 |
| Google Syndication | Alphabet* | 24.6 | 9.7 | 32.3 | 32.0 |
| AppNexus | AT&T | 24.3 | 12.7 | 29.7 | 43.2 |
| Yandex | Yandex | 14.1 | 6.3 | 24.2 | 4.7 |
| Criteo | Criteo* | 19.3 | 10.7 | 23.5 | 30.8 |
| Amazon Advertising | Amazon* | 15.7 | 7.7 | 22.5 | 21.0 |
| Quantcast | Quantcast | 12.4 | 8.0 | 18.7 | 7.6 |
| Adobe Audience Manager | Adobe | 14.4 | 8.0 | 18.5 | 20.3 |
| Bing Ads | Microsoft* | 17.2 | 9.6 | 16.6 | 22.7 |
| TradeDesk | The Trade Desk | 10.7 | 2.5 | 14.7 | 5.9 |
| MediaMath | MediaMath | 8.9 | 2.7 | 13.6 | 7.5 |
| Moat | Oracle | 4.2 | 1.4 | 13.2 | 1.0 |
| *B. Combined provider coverage* | | | | | |
| Any advertising-related technology | | 94.8 | 77.7 | 96.9 | 93.5 |
| Liable-owned providers | | 88.9 | 67.9 | 92.9 | 90.6 |
| Nonliable-owned providers | | 76.2 | 52.2 | 79.8 | 70.9 |
| Unresolved ownership | | 15.2 | 4.7 | 32.4 | 26.1 |

*Notes:* All values are percentages. Both panels restrict technologies to WTM's advertising category. The universe consists of all 638 nonliable website families observed in November 2018 either in the French traffic extract or the global traffic extract. A published detection on any member domain identifies a website-level connection. Panel A is ranked by global nonliable traffic coverage. Panel B uses all advertising-related technologies, including those outside the top fifteen, counting a family once per row. Rows overlap and must not be added. Corporate owners are as of January 2019; * denotes an owner classified as likely liable for the French DST in this study. Source: WhoTracksMe site and site–tracker panels (Karaj et al., 2018); saved April 2026 WTM category metadata for advertising categorization.

# 3 Digital Tax Incentives and Website Popularity

This section connects the economic incentives created by the digital services tax to observed website traffic. The central idea is that the tax can change platform choices that are unobserved in the data but affect users' willingness to visit. This connection is first described without specifying how platforms produce or monetize their services. Two parsimonious models then provide frameworks for interpreting the traffic response. The broad claim that traffic responses signal tax-induced platform adjustments does not depend on the specific assumptions of the models below. The models clarify possible economic mechanisms and, under their maintained assumptions, allow the traffic estimates to be combined with price evidence to decompose the responses into distinct channels.

Let $\mathbf{q}_i$ denote a vector of characteristics of website $i$ that result from platform choices. They may include the content available, the convenience of accessing it, advertising intensity, and other amenities or disamenities. Users' choices about how often to visit therefore reveal their demand for the bundle of characteristics the platform provides.

$$\mathcal{L}_i(\mathbf{q}_i) = \arg\max_{L_i \geq 0} v_i(L_i; \mathbf{q}_i). \tag{4}$$

Here, $v_i$ denotes the user's utility from visiting website $i$, given its characteristics. This formulation allows a site-specific satiation point without introducing an explicit consumption cost.

When a new digital service tax $\tau$ distorts the platform's choices, the user's demand for page loads changes according to:

$$\frac{\partial \mathcal{L}_i}{\partial \tau} = \sum_k \frac{\partial \mathcal{L}_i}{\partial q_{ik}} \frac{dq_{ik}}{d\tau}. \tag{5}$$

The response combines two elements: how the platform changes each characteristic and how users' demand responds to that characteristic. This relationship does not require a particular model of platform production or monetization.

If the tax affects a website's traffic only through its own characteristics, a nonzero causal effect on traffic implies that the tax induced a change in at least one platform choice that matters for user demand. A zero effect admits several interpretations. The platform may leave its choices unchanged; users' visits may respond little to the characteristics that change; or the traffic effects of different adjustments may offset one another. Traffic alone therefore cannot identify which choices changed or establish that the platform did not respond.

The next two subsections illustrate this argument using models of an advertising-funded social media platform and an e-commerce platform that intermediates third-party transactions. These models connect the two statutory tax bases to possible platform responses. They provide interpretations of the estimated traffic effects rather than claims about the particular adjustments platforms made.

To focus on domestic distortions, it is assumed that $\bar{p} = p_F$ in Equation 2, so that the effective marginal

tax rate on French activity equals the statutory rate of 3%. The models below abstract from adjustments in foreign activity.

### 3.1 Social Media Platforms

Now consider the promised hypothetical social media platform that invests in engaging content to attract users and monetizes the resulting attention by showing them advertisements. The platform incurs two types of expenditure: $g_U$ funds services that attract users, while $g_A$ funds the sale and delivery of advertising. A tax on advertising revenue reduces the return to both. Its effect on traffic depends on how the platform adjusts the services it provides and the advertising users encounter.

Let $L$ denote page loads and $A$ advertising impressions. Advertising intensity, $a = A/L$, measures impressions per page load. Traffic satisfies

$$L = \mathcal{L}(a, g_U), \qquad \mathcal{L}_a < 0, \quad \mathcal{L}_{g_U} > 0, \quad \mathcal{L}_{g_U g_U} \leq 0.$$

User-facing expenditure attracts traffic with diminishing returns. Advertising discourages visits because it interrupts or competes with the content users seek. This assumption describes a content-consumption setting; advertisements that help users find products can have a different effect.

Delivering impressions requires advertiser-facing expenditure according to

$$A = F(g_A), \qquad F'(g_A) > 0, \quad F''(g_A) \leq 0.$$

The platform faces inverse advertising demand $p_A = P_A(A)$, with $P_A'(A) < 0$. Advertisers pay less per impression as the platform supplies more impressions. Impression quality and audience composition are held fixed, so their willingness to pay depends on the number of impressions rather than separately on advertising intensity or user-facing expenditure. Appendix C.1 derives this demand curve from a downstream product market.

Let $\tau \in [0, 1)$ denote a tax on advertising revenue. The platform chooses advertising intensity and both expenditures to maximize

$$\begin{aligned} \max_{a, g_U, g_A \geq 0} \quad & (1-\tau) P_A(A) A - g_A - g_U \\ \text{subject to} \quad & A = F(g_A) = a\mathcal{L}(a, g_U). \end{aligned}$$

This characterization of advertising production serves as a realistic portrait of the toy social media platform. Note that the characterization of the firm's costs is meant to emphasize the idea that advertising and traffic are best thought of as downstream outcomes of ex-ante investment decisions made by digital platforms. Doing so helps to avoid confusion about the marginal cost structure of digital production.

Although for pedagogical reasons it is useful to characterize traffic and advertising as an outcome of unobservable platform choices, in practice it is mathematically convenient to equivalently express the optimization problem simply as a function of $A$. To do so, start by noting that for a given number of impressions,

revenue is fixed. The platform therefore chooses the least costly combination of advertising intensity and user-facing expenditure that supports those impressions. Define this user-facing expenditure as

$$U(A) = \min_{a, g_U \geq 0} \{g_U : \ a\mathcal{L}(a, g_U) = A\}.$$

Advertiser-facing expenditure is $F^{-1}(A)$, so total required expenditure is $C_A(A) = F^{-1}(A) + U(A)$. This characterization of the cost function is especially useful conceptually, as it emphasizes that users are a costly input into the production of advertising.

At the expenditure-minimizing allocation, the first-order and envelope conditions derived in Appendix C.2 give

$$\varepsilon_{L,a} \equiv \frac{a\mathcal{L}_a}{\mathcal{L}} = -1, \qquad U'(A) = \frac{1}{a\mathcal{L}_{g_U}} > 0. \tag{6}$$

The first condition balances the extra impressions from showing another advertisement on each page against the impressions lost when users visit less. At the maintained interior optimum, users are therefore not indifferent to advertising. The second states that supporting more impressions requires more expenditure on the services that attract users. A zero total traffic response to the tax instead reflects offsetting changes in advertising intensity and user-facing expenditure.

The platform's problem of how many impressions to deliver can then be written as

$$\max_{A \geq 0} \Pi(A; \tau), \qquad \Pi(A; \tau) = (1 - \tau) P_A(A) A - C_A(A),$$

over feasible volumes with nonnegative advertising prices. Its first-order condition is

$$(1 - \tau) \left[ P_A(A) + A P'_A(A) \right] = C'_A(A).$$

This expression states the optimality condition that after-tax marginal advertising revenue equals the marginal expenditure needed to supply another impression. A higher tax lowers this return and reduces the chosen volume. This yields the first comparative static expression for total advertising:

$$\frac{dA}{d\tau} = \frac{P_A(A) + A P'_A(A)}{\Pi_{AA}(A; \tau)} < 0.$$

The numerator is positive because marginal expenditure is positive; the denominator is negative by the strict second-order condition. Fewer impressions require less advertiser-facing expenditure and less user-facing expenditure; Appendix C.2 derives these tax responses. In particular,

$$\frac{dg_U}{d\tau} = \frac{1}{a\mathcal{L}_{g_U}} \frac{dA}{d\tau} < 0. \tag{7}$$

Because of the decline in both ads and user-facing investments, the decline in impressions need not mean

fewer advertisements per page. Appendix C.3 derives the response of advertising intensity to the reduction in user-facing expenditure:

$$\frac{da}{d\tau} = -\frac{\mathcal{L}_{g_U} + a\mathcal{L}_{ag_U}}{2\mathcal{L}_a + a\mathcal{L}_{aa}} \frac{dg_U}{d\tau}.$$

The denominator is negative at the conditional expenditure minimum. Intensity therefore falls when $\mathcal{L}_{g_U} + a\mathcal{L}_{ag_U} > 0$: greater expenditure on user services raises the marginal impression gain from increasing advertising intensity, so reducing that expenditure makes lower intensity optimal. If this expression is negative, intensity rises.

Traffic combines the effects of these two adjustments:

$$\frac{dL}{d\tau} = \underbrace{\mathcal{L}_a \frac{da}{d\tau}}_{\text{advertising intensity}} + \underbrace{\mathcal{L}_{g_U} \frac{dg_U}{d\tau}}_{\text{user services}}. \tag{8}$$

At the interior optimum, the tax reduces user-facing expenditure. To see when traffic nevertheless rises, consider how much the platform would have to reduce ads per page to keep traffic constant after the cut in content spending in Equation (7). Along a constant-traffic curve, $\mathcal{L}_a\, da + \mathcal{L}_{g_U}\, dg_U = 0$, so

$$\left.\frac{da}{dg_U}\right|_L = -\frac{\mathcal{L}_{g_U}}{\mathcal{L}_a} > 0.$$

This slope measures how much ad intensity must fall per unit reduction in user-facing expenditure to keep page loads unchanged. Traffic rises if the platform cuts ad intensity by more than this benchmark, falls if it cuts by less, and is locally unchanged if the two adjustments coincide.

In elasticity terms, let $\varepsilon_{L,g_U}$ and $\varepsilon_{L,a}$ denote the partial elasticities of traffic with respect to user-facing expenditure and ad intensity. At the expenditure-minimizing allocation, Equation (6) gives $\varepsilon_{L,a} = -1$, so the condition for traffic to rise is

$$\frac{dL}{d\tau} > 0 \iff \underbrace{\frac{d\log a/d\tau}{d\log g_U/d\tau}}_{\text{firm's optimal adjustment}} > \underbrace{\frac{\varepsilon_{L,g_U}}{-\varepsilon_{L,a}}}_{\text{constant-traffic tradeoff}} = \varepsilon_{L,g_U}.$$

For a 1 percent cut in user-facing expenditure, traffic rises if the percentage cut in ads per page exceeds the percentage of traffic that the spending cut would have cost at fixed ad intensity.

To see why the firm may cut ad intensity by more or less than the constant-traffic benchmark, consider its incentive at that benchmark after the reduction in user-facing expenditure from Equation (7). If users there are more than unit elastic with respect to ad intensity, the impressions lost when users load fewer pages outweigh the extra impressions on the pages that remain. The firm can therefore gain impressions by cutting ad intensity further, which raises traffic above its initial level. If users are less than unit elastic, the firm can gain impressions by showing more ads per page than at the benchmark, and traffic falls. If the elasticity is exactly minus one, the benchmark already satisfies the firm's optimality condition.

The model also links the decline in impressions $A$ to the traffic loss from reduced content and services. Using $A = a\mathcal{L}(a, g_U)$, a first-order approximation around the initial optimum gives

$$\Delta \log A \approx \underbrace{(1 + \varepsilon_{L,a})}_{=0} \Delta \log a + \varepsilon_{L,g_U} \Delta \log g_U = \varepsilon_{L,g_U} \Delta \log g_U.$$

Here $\varepsilon_{L,g_U}$ is the traffic elasticity with respect to user-facing expenditure at fixed intensity. The intensity term vanishes by the envelope argument for expenditure minimization. The optimality condition $\varepsilon_{L,a} = -1$ in Equation (6) means that changing intensity has no first-order effect on impressions at fixed spending. Thus, to first order, the log decline in impressions equals the log traffic loss from reduced content and services at fixed intensity.

When the platform shows fewer ads per page at fixed content spending, it gives up impressions on pages users would have loaded anyway and gains impressions on the extra pages they now choose to load. At the optimum these effects cancel to first order, even as page loads rise. Cutting content reduces both page loads and the impressions those pages would have carried at the prevailing ad intensity. If fewer ads per page bring back more page loads than the content cut takes away, the platform shows fewer impressions across more pages.

### 3.2 Transaction Intermediation

Now, consider the effects of digital taxation on a hypothetical e-commerce intermediator. Relative to the social media case, this setting emphasizes two channels through which the tax can affect traffic: changes in the prices buyers pay and changes in the sellers available on the platform.

Let $Q$ denote completed transactions, $p_B$ the price paid by buyers, and $p_S$ the amount sellers receive per transaction. Buyer demand and aggregate seller supply imply

$$p_B = D^{-1}(Q), \qquad p_S = S^{-1}(Q),$$

with $(D^{-1})'(Q) < 0$ and $(S^{-1})'(Q) > 0$. The platform's fee is the difference between these prices:

$$p_Q = P_Q(Q) \equiv D^{-1}(Q) - S^{-1}(Q).$$

Higher fees reduce transaction volume: $P_Q'(Q) < 0$. The division of the fee between higher buyer prices and lower seller compensation depends on the slopes of demand and supply.

To distinguish availability from completed transactions, let

$$N = H(p_S), \qquad H'(p_S) > 0,$$

denote the number of active sellers. Higher compensation makes participation worthwhile for more sellers,

expanding the set of offers available to users. Aggregate supply $S(p_S)$ incorporates both seller participation and the quantities supplied by participating sellers. The participation response is an additional assumption; upward-sloping aggregate supply alone does not imply it.

Intermediation expenditure, $g_Q$, funds transaction processing, buyer and seller support, and other resources needed to facilitate purchases. Supporting transactions requires

$$Q = F(g_Q), \qquad F'(g_Q) > 0, \quad F''(g_Q) \leq 0.$$

The required expenditure is therefore $C_Q(Q) = F^{-1}(Q)$, with $C_Q'(Q) > 0$. The model treats this expenditure as a cost of supporting transaction volume. Its effects on traffic operate through the buyer prices and seller participation associated with that volume.

Let $\tau \in [0, 1)$ denote a tax on fee revenue. The platform chooses transaction volume to maximize

$$\max_{Q \geq 0} \Pi(Q; \tau), \qquad \Pi(Q; \tau) = (1 - \tau) P_Q(Q) Q - C_Q(Q), \tag{9}$$

over feasible volumes with nonnegative fees. The taxable revenue is the platform's fee income, $p_Q Q$. Smooth, positive interior solutions satisfying the strict second-order condition are considered. Appendix D.1 derives the expenditure conditions and transaction-volume response.

The platform equates after-tax marginal fee revenue with the marginal expenditure required to support another transaction:

$$(1 - \tau) \left[ P_Q(Q) + Q P_Q'(Q) \right] = C_Q'(Q).$$

A higher tax lowers the return to additional transactions while leaving their expenditure requirements unchanged. The chosen volume falls:

$$\frac{dQ}{d\tau} = \frac{P_Q(Q) + Q P_Q'(Q)}{\Pi_{QQ}(Q; \tau)} < 0.$$

The numerator is positive by the first-order condition, since marginal expenditure is positive. The denominator is negative by the strict second-order condition.

Lower transaction volume requires less intermediation expenditure, as derived in Appendix D.1, and corresponds to a higher fee. Appendix D.2 derives the fee and price responses. Along buyer demand and seller supply,

$$\frac{dp_B}{d\tau} = (D^{-1})'(Q) \frac{dQ}{d\tau} > 0, \qquad \frac{dp_S}{d\tau} = (S^{-1})'(Q) \frac{dQ}{d\tau} < 0.$$

Buyers pay more and sellers receive less. Lower seller compensation also reduces participation:

$$\frac{dN}{d\tau} = H'(p_S) \frac{dp_S}{d\tau} < 0.$$

Thus, the tax reduces both completed transactions and the number of sellers available to users; Appendix D.2

derives the participation response under the separate assumption $H' > 0$.

Traffic includes browsing and comparison shopping as well as completed purchases. Let $L$ denote page loads. Traffic depends on buyer prices and seller availability:

$$L = \widetilde{\mathcal{L}}(p_B, N).$$

Assume that higher buyer prices discourage visits and greater seller availability attracts them:

$$\widetilde{\mathcal{L}}_{p_B} < 0, \qquad \widetilde{\mathcal{L}}_N > 0.$$

These are assumptions about the net response of page loads, including visits that do not result in a purchase. In particular, the second assumption treats seller participation as an amenity for users, rather than equating availability with the number of completed transactions.

The traffic response is

$$\frac{dL}{d\tau} = \underbrace{\widetilde{\mathcal{L}}_{p_B} \frac{dp_B}{d\tau}}_{\text{buyer prices}} + \underbrace{\widetilde{\mathcal{L}}_N H'(p_S) \frac{dp_S}{d\tau}}_{\text{seller availability}} < 0.$$

Higher buyer prices discourage visits, while reduced seller participation makes the platform less attractive for browsing and purchasing. Both channels lower traffic under the maintained assumptions.

The negative traffic prediction depends on these responses. Appendix D.2 derives the more general condition: traffic falls when

$$\widetilde{\mathcal{L}}_{p_B}(D^{-1})'(Q) + \widetilde{\mathcal{L}}_N H'(p_S)(S^{-1})'(Q) > 0,$$

so that traffic increases with transaction volume along the market equilibrium. The sign assumptions above are sufficient for this condition. If higher prices or reduced availability instead lead users to search more intensively, page loads need not fall with transactions.

# 4 Estimating Traffic Responses

The core of this paper's empirical strategy consists of estimating the response to the French digital services tax by comparing changes in French-origin traffic between websites owned by liable firms and nonliable websites. The comparison asks whether traffic to liable websites changes after the December 2018 announcement relative to traffic to websites outside the tax's scope. The comparison covers September–November 2018 and January–December 2019, excluding December 2018 because the announcement occurs during that month. Ending the comparison in 2019 avoids combining the initial response with changes in internet use during the COVID-19 pandemic.

The preferred specification compares liable website families with synthetic controls constructed from nonliable families without observed advertising exposure. Additional comparisons use advertising-exposed

and pooled nonliable websites to assess how the choice of comparison group affects the estimates.

The baseline difference-in-differences specification is

$$y_{it} = \alpha_i + \lambda_t + \beta \left( L_i \times Post_t \right) + \varepsilon_{it}, \tag{10}$$

where $i$ indexes website families and $t$ indexes calendar months. The outcome $y_{it}$ is the logarithm of family $i$'s French-origin traffic relative to Wikipedia in month $t$. The indicator $L_i$ identifies families owned by liable firms, and $Post_t$ equals one in 2019. Website-family fixed effects, $\alpha_i$, absorb persistent differences in traffic across families, while month fixed effects, $\lambda_t$, absorb monthly changes common to the sample. The coefficient $\beta$ measures the change in log traffic to liable websites relative to nonliable websites.

The regression uses ordinary least squares, giving each observed family-month equal weight. The estimate therefore describes differential traffic growth across website families rather than the change in their combined traffic volume. The bootstrap calculates standard errors by resampling whole parent firms for liable websites and whole website families for nonliable websites, retaining all websites and months belonging to each sampled block. This allows dependence across liable websites under the same parent and over time within each block. The full-pipeline procedure below reestimates the synthetic weights in every draw.

The spillover specification examines potential effects within the nonliable group using the presence of advertising-related tracking technologies. The spillover comparison contrasts traffic changes between nonliable website families with and without observed advertising exposure, with advertising exposure defined in Subsection 4.3:

$$y_{it} = \alpha_i + \lambda_t + \delta \left( E_i \times Post_t \right) + \varepsilon_{it}, \tag{11}$$

where the sample contains only nonliable website families and $E_i$ indicates an observed connection to a liable-owned WTM advertising technology in the French panel during September–November 2018. The estimation period and weighting follow the baseline specification; inference resamples full family histories separately within the exposed and residual groups.

The coefficient $\delta$ measures the post-announcement change in log traffic among nonliable families with observed exposure relative to those without observed exposure. A causal spillover interpretation requires that these groups would have experienced parallel traffic trends absent the tax and that the comparison group is unaffected. For the main synthetic comparison, a separate donor vector is fitted for each of the 469 advertising-exposed nonliable families, drawing from the unexposed group. This comparison uses the French-only fitting criterion described below, with French-only donor eligibility and no French-audience exposure balancing term. Figure 5 and Table 8 report the resulting paths and period estimates.

The dynamic specification extends Equation 10 as follows:

$$y_{it} = \alpha_i + \lambda_t + \sum_{m \in \mathcal{T} \setminus \{\text{Nov. 2018}\}} \beta_m \left( L_i \times \mathbf{1}\{t = m\} \right) + \varepsilon_{it}, \tag{12}$$

where $\mathcal{T}$ contains the estimation months and November 2018 is the omitted reference month. Each coefficient $\beta_m$ measures the liable–nonliable difference in log traffic in month $m$ relative to that difference in November 2018. For presentation, the normalization centers the coefficients on their September–November 2018 average:

$$\widetilde{\beta}_m = \beta_m - \frac{1}{3}\sum_{s\in\mathcal{P}} \beta_s, \qquad \mathcal{P} = \{\text{Sep., Oct., Nov. 2018}\}, \tag{13}$$

with $\beta_{\text{Nov. 2018}} = 0$. The dynamic estimates show how the relative traffic response evolves during 2019 and whether differential changes are visible within the baseline period. Standard errors and pointwise percentile intervals use the full-history bootstrap described below.

A causal interpretation requires that, absent the tax announcement, liable and nonliable websites would have experienced parallel changes in log traffic. It also requires that the tax does not affect the comparison websites. This second condition deserves particular attention because nonliable websites can interact with liable platforms through advertising and other services. If these relationships transmit the tax's effects, the estimate captures the response of liable websites relative to an affected comparison group.

The corresponding dynamic specification among nonliable website families replaces $(L_i)$ with the advertising-exposure indicator $(E_i)$. Its monthly coefficients measure exposed–unexposed differences relative to November 2018. For presentation, the normalization centers these coefficients on their September–November 2018 average, using the same normalization as above.

The conceptual framework in Section 3 identifies competing effects of the tax on advertising-supported platforms. A reduction in advertising intensity can increase traffic, while a reduction in user-facing expenditure can decrease it. The advertising-share interaction measures how traffic responses vary with reliance on advertising.

The specification measures advertising's share of in-scope consolidated revenue, defined as advertising revenue divided by the sum of advertising and intermediation revenue:

$$y_{it} = \alpha_i + \lambda_t + \beta\,(L_i \times Post_t) + \theta_C \left(L_i \times Post_t \times \widetilde{C}_{j(i)}\right) + \varepsilon_{it}, \tag{14}$$

where $j(i)$ denotes the parent firm of website family $i$ and $C_j = R_j^A/(R_j^A + R_j^I)$. The variables $R_j^A$ and $R_j^I$ denote consolidated advertising and intermediation revenue as reported in Table 2. This ratio describes the composition of revenue from the two in-scope activities, rather than their importance in the firm's total revenue.

Fiscal-year 2018 accounts provide $C_j$, which remains fixed throughout the analysis. The standardized measure $\widetilde{C}_j$ subtracts its mean and divides by its standard deviation among the 55 included liable website families. The coefficient $\beta$ compares liable and nonliable traffic changes at mean advertising composition; $\theta_C$ is the gradient per one-standard-deviation increase in composition. The estimation period, equal family-month weights, fixed effects and bootstrap resampling blocks follow the baseline specification. For synthetic comparisons, the specification uses the same revenue interaction on the liable member and retains member

and shared month fixed effects.

A causal interpretation of this gradient requires that, absent the tax announcement, traffic changes among liable websites would not have varied systematically with advertising composition.

The results include both the coefficient estimates and the implied traffic responses for hypothetical pure advertising and pure intermediation firms. The coefficients describe how the response varies with revenue composition, while the hypothetical profiles express these relationships as predicted liable–nonliable differences in traffic changes.

For Equation 14, a pure advertising firm has $C_j = 1$ and a pure intermediation firm has $C_j = 0$. Their implied responses are

$$\Delta_A^C = \beta + \theta_C \frac{1 - \overline{C}}{s_C}, \tag{15}$$

$$\Delta_I^C = \beta + \theta_C \frac{-\overline{C}}{s_C}, \tag{16}$$

where $\overline{C}$ and $s_C$ are the mean and standard deviation used to standardize $C_j$. Here, "pure" refers to the composition of in-scope revenue; either firm may also earn revenue from other activities.

Each bootstrap draw evaluates the full expression for each hypothetical profile. Standard errors are the standard deviation of the resulting estimates, retaining dependence between the intercept and composition slope; confidence intervals are their 2.5th and 97.5th percentiles.

### 4.1 Empirical Bayes estimates of average website effects.

The hypothetical profiles above describe predictions from the composition model. An Empirical Bayes shrinkage strategy estimates average effects among observed pure advertising and pure intermediation websites while allowing individual websites to depart from that relationship. The calculation uses all 55 liable families and their unexposed synthetic comparisons. The calculation begins with each family's raw annual contrast

$$\widehat{\delta}_i = \frac{1}{12} \sum_{t \in \mathcal{Q}} g_{it} - \frac{1}{3} \sum_{t \in \mathcal{P}} g_{it}, \qquad \mathcal{Q} = \{\text{Jan.–Dec. 2019}\}, \tag{17}$$

where $g_{it}$ is the observed-minus-synthetic log traffic gap and $\mathcal{P}$ contains September–November 2018. December 2018 is excluded. The composition model supplies the center

$$m_i = \widehat{\beta} + \widehat{\theta}_C \frac{C_{j(i)} - \overline{C}}{s_C}, \tag{18}$$

using the original stacked synthetic-control specification in Equation (14). This center is not separately refitted by regressing website gaps on composition.

The working empirical Bayes model is

$$\widehat{\delta}_i \mid \delta_i \sim N(\delta_i, s_i^2), \qquad \delta_i \mid C_{j(i)} \sim N(m_i, \tau^2), \tag{19}$$

where $s_i^2$ is the estimated sampling variance of the raw contrast and $\tau^2$ measures dispersion of website effects around the composition prediction. To calibrate $s_i^2$, each website's monthly gap over the three baseline months and twelve months of 2019 is fitted with a website-specific baseline and a constant website-specific post-announcement shift. Residuals follow a common AR(1) correlation structure with website-specific variance scales. Contrast variances account for baseline covariance and actual calendar distances, including the gap between November and January. December is excluded from this calibration. This working noise model treats fluctuations around a website's constant 2019 shift as noise; genuine dynamic responses or synthetic mismatch can therefore contribute to the estimated sampling variance.

Holding the fitted centers fixed during the variance optimization, the nonnegative residual variance $\tau^2$ is estimated by maximizing the plug-in Gaussian marginal likelihood of $\widehat{\delta}_i$ with mean $m_i$ and variance $\tau^2 + s_i^2$. The resulting website estimate is

$$\widetilde{\delta}_i = r_i \widehat{\delta}_i + (1 - r_i) m_i, \qquad r_i = \frac{\widehat{\tau}^2}{\widehat{\tau}^2 + \widehat{s}_i^2}. \tag{20}$$

More precise raw contrasts receive greater weight; less precise contrasts shrink toward the composition model. The notation $r_i$ distinguishes this reliability weight from synthetic donor weights. Group averages give equal weight to $\widetilde{\delta}_i$ across the 13 observed pure advertising families ($C = 1$, five parents) and, separately, the five pure intermediation families ($C = 0$, five parents). These observed-group averages differ from the hypothetical model-only profiles $\Delta_A^C$ and $\Delta_I^C$. All reported effects are multiplied by 100, without exponentiation.

Annual raw-group and EB standard errors and percentile confidence intervals use 1,999 full-pipeline bootstrap draws. Parent sampling is stratified by the three fixed revenue-composition groups: five advertising parents, ten mixed parents and five intermediation parents are sampled with replacement within their groups. Each selected parent's entire set of website families and months is retained. The 169 residual nonliable family histories are sampled separately. Each draw reconstructs synthetic paths, reestimates the original stacked-model centers, AR(1) correlation, website noise scales, residual signal variance, reliability weights and group averages. Within a draw, the centers are treated as given only during residual-variance optimization; they are reestimated across draws. Inference therefore incorporates donor-sample and synthetic-weight variation as well as EB parameter estimation.

The dynamic calculation repeats the same construction separately for each month using the original monthly stacked-model predictions, month-specific raw gap contrasts, and sampling variances from the same AR(1) calibration. The residual signal variance is estimated separately for each month. The normalization centers fitted website paths on their own September–November 2018 means, and group averages

give each website equal weight within each reported group. December 2018 enters as a separate transition month, while remaining excluded from the annual contrast and noise calibration. Pointwise 95% percentile intervals use the same 1,999 revenue-stratified full-pipeline draws, repeating synthetic fitting, monthly signal estimation and normalization in every draw. These monthly fits do not impose a joint Bayesian dynamic model. The annual empirical Bayes estimate is constructed from annual contrasts and need not equal the average of the separately fitted monthly estimates.

Appendix F reports the specifications and results for the supplementary French-share analysis (Figure F.1 and Table F.1) and joint revenue-share analysis (Figure F.2 and Table F.2).

The following subsections describe sample construction and the characteristics of nonliable websites, including their connections to liable platforms.

## 4.2 Data and Sample Construction

A monthly panel of website-family level data on French-origin web traffic supports this estimation strategy. Its construction proceeds as follows.

WhoTracksMe reports monthly website popularity relative to Google within each geographic market. The original measure therefore reflects a website's traffic relative to Google's traffic, rather than an absolute traffic count. Let $p_{dFt}$ denote this Google-normalized popularity for domain $d$, among users in France, in month $t$.

Summing domain popularity first aggregates domains into website families:

$$P_{it}^{F} = \sum_{d \in \mathcal{D}_i} p_{dFt},$$

where $\mathcal{D}_i$ contains the domains belonging to family $i$. Because the member domains share the same Google denominator, their sum expresses family traffic relative to Google.

Dividing family popularity by Wikipedia's Google-normalized popularity in the same month and geographic market then expresses family popularity relative to Wikipedia:

$$y_{it} = \log\left(\frac{P_{it}^{F}}{p_{\text{Wikipedia},Ft}}\right).$$

The original Google denominator cancels in this ratio, leaving the logarithm of family traffic relative to Wikipedia.

The choice of reference website does not affect within-month treated–control log differences, because the common reference term also cancels:

$$y_{it} - y_{jt} = \log P_{it}^{F} - \log P_{jt}^{F}.$$

The same cancellation holds for a given synthetic comparison whose weights sum to one. In the difference-

in-differences regressions, month fixed effects absorb the common monthly reference term.

The panel includes only website families with sufficient observations before the December 2018 announcement to assess pre-treatment traffic patterns. For families owned by liable corporations, panel eligibility generally requires at least four months of observations between June and November 2018. Futhead is retained as an exception, with observations for September–November 2018. For nonliable families, panel eligibility requires observations in all six months. This stricter coverage requirement addresses the sharp expansion in the number of websites published by WTM during the pre-announcement period, which leaves many potential comparison websites with short traffic histories.

The resulting panel contains 55 liable website families belonging to 20 parent firms (Table 3) and 638 nonliable website families. The panel retains available observations during January–December 2019 without requiring complete post-announcement coverage. The estimation baseline is September–November 2018; December 2018 is excluded from the annual comparisons because the announcement occurs during that month.

The heterogeneity specifications attach parent-level revenue measures to each liable website family. The French audience concentration measure is instead constructed at the family level using pre-announcement French and global popularity. These characteristics are held fixed over time. The following subsections describe the advertising connections used to define comparison groups and the synthetic-control construction.

### 4.3 Advertising Exposure among Nonliable Websites

The advertising connections described in Subsection 2.4 can transmit the DST to nonliable websites. This section describes how historical data on the presence of advertising-related tracking technologies link third party sites to advertising activity related to the liable platforms in this study.

The starting point is the WhoTracksMe dataset, which records detections of web technologies on individual website domains in its country-specific monthly panels. These technologies include advertising services, analytics tools, and other website infrastructure. For each detected technology, WTM provides an identity and category classification. Detections in its `advertising` category in the French panel measure observed advertising connections among nonliable websites.

Table 6 reports observed advertising connections among the 638 nonliable website families with balanced pre-treatment traffic observations. Exposure requires a detection on at least one member domain during September–November 2018. The table distinguishes connections to any advertising provider from connections to providers owned by liable corporations.

These measures identify technical connections rather than advertising expenditures, receipts, or contractual relationships. An observed connection does not establish whether a website buys or sells advertising, how much revenue is involved, or whether the tax is passed through to it. Conversely, the absence of a qualifying observation does not establish the absence of advertising relationships. The residual category therefore carries the label "no observed exposure," including families whose exposure is unknown.

This exercise identifies 169 website families without an observed advertising connection to liable platforms. These families form the starting pool from which eligible synthetic-control donors are selected. The remaining 469 families have observed connections to liable advertising providers; a separate comparison of their traffic estimates spillover effects on nonliable websites.

Table 6: Observed advertising exposure among nonliable website families

*September–November 2018*

| Exposure measure | Observed | Not observed |
|---|---|---|
| Any-provider advertising | 531 | 107 |
| Liable advertising | 469 | 169 |

*Notes:* The table classifies 638 nonliable website families using positive published detections in the French WTM panel and identities in its `advertising` category. A detection on any member domain in any month of September–November 2018 identifies exposure for the family. Any-provider advertising includes providers regardless of liability; liable advertising includes only providers owned by firms classified as liable in this study. The main measure is the liable-provider row: 469 observed and 169 not observed. Rows overlap and should not be added. Categories use the saved April 2026 metadata and liability uses the study crosswalk.

## 4.4 Constructing a Panel of Synthetic Controls

The preferred estimation strategy uses stacked synthetic controls. For each of the 55 liable website families and 469 advertising-exposed nonliable families, a synthetic comparison is constructed from nonliable families without observed pre-treatment advertising connections to liable platforms.

Among the 169 eligible sites without observed liable-tech exposure, donor eligibility further requires full balance for the entire analysis period. This results in 159 donors.

For each target family $i$, the French synthetic path is

$$\widehat{y}_{it}^{0} = a_i + \sum_{j \in \mathcal{D}_i} w_{ij} y_{jt}, \qquad w_{ij} \geq 0, \qquad \sum_{j \in \mathcal{D}_i} w_{ij} = 1. \tag{21}$$

Each French target and donor history is demeaned over the target's observed June–November 2018 training months. The mean squared trajectory loss is divided by the variance of donor traffic changes over those months, and a ridge penalty with coefficient $\sqrt{12}$ limits weight concentration. Weights are nonnegative and sum to one. The intercept $a_i$ ensures that the synthetic path matches the target's average French traffic over September–November 2018.

Let $\mathcal{T}_i$ denote the target's observed June–November 2018 months, and let $\widetilde{y}_{kt}^{(i)}$ demean each history over those same months. Then:

$$\widehat{\mathbf{w}}_i = \underset{\mathbf{w}_i \geq 0,\ \mathbf{1}'\mathbf{w}_i = 1}{\arg\min} \left\{ \frac{1}{|\mathcal{T}_i| \widehat{\sigma}_i^2} \sum_{t \in \mathcal{T}_i} \left( \widetilde{y}_{it}^{(i)} - \sum_{j \in \mathcal{D}_i} w_{ij} \widetilde{y}_{jt}^{(i)} \right)^2 + \sqrt{12} \sum_{j \in \mathcal{D}_i} w_{ij}^2 \right\}.$$

Here, $\widehat{\sigma}_i^2$ is the sample variance of donor month-to-month traffic changes, pooled across donors and consec-

utive months observed for the target. The first term rewards matching the pre-announcement trajectory; the second discourages concentrating weight on a few donors.

To estimate the weights, pre-announcement traffic histories form a feature vector $\mathbf{x}_i(\mathcal{F})$, where $\mathcal{F}$ denotes the set of countries whose traffic histories enter the fitting criterion. The candidate fitting procedures use French traffic alone or French traffic combined with German and US traffic. For each included country, demeaning uses the target's observed training months and centers each target and donor history over those same months. Scaling divides each country's demeaned histories by $\sqrt{n_{ic}}\widehat{\sigma}_{ic}$, where $n_{ic}$ is the number of observed training months for target $i$ in country $c$, and $\widehat{\sigma}_{ic}$ is the standard deviation of donor traffic changes, pooled across eligible donors and consecutive observed training months. This scaling expresses discrepancies relative to donor traffic fluctuations and prevents countries with more observed months from mechanically receiving greater weight.

Let $\mathbf{x}_j^{(i)}(\mathcal{F})$ denote the corresponding feature vector for donor $j$, constructed using target $i$'s training months and scaling factors. Solving the following minimization problem yields the weights:

$$\widehat{\mathbf{w}}_i(\mathcal{F}) = \underset{\substack{w_{ij}\geq 0 \\ \sum_{j\in\mathcal{D}_i(\mathcal{F})} w_{ij}=1}}{\arg\min} \left\{ \left\| \mathbf{x}_i(\mathcal{F}) - \sum_{j\in\mathcal{D}_i(\mathcal{F})} w_{ij}\mathbf{x}_j^{(i)}(\mathcal{F}) \right\|_2^2 + \lambda \sum_{j\in\mathcal{D}_i(\mathcal{F})} w_{ij}^2 \right\}. \tag{22}$$

The first term rewards matching the target's pre-announcement traffic trajectory. The second discourages concentrated weights. The regularization coefficient remains fixed at $\lambda = \sqrt{12}$ across specifications; the validation exercise below selects between fitting procedures rather than tuning this coefficient. Once the weights are estimated using all available June–November 2018 training observations, the intercept aligns the synthetic path with the target's September–November 2018 mean:

$$\widehat{a}_i = \frac{1}{3} \sum_{t\in\{\text{Sep.,Oct.,Nov. 2018}\}} \left( y_{it} - \sum_{j\in\mathcal{D}_i} \widehat{w}_{ij} y_{jt} \right). \tag{23}$$

The validation exercise assesses the value of adding foreign traffic histories using three forward validation splits within the pre-announcement period. The first fits June–August 2018 and predicts September–November; the second fits June–September and predicts October–November; the third fits June–October and predicts November. The evaluation samples contain 49, 54 and 55 liable website families, respectively. Within each split, training observations alone determine all demeaning transformations, donor-change scales and weights. For prediction, the French intercept equals the mean observed-minus-weighted-donor gap in the training months, without recentering on held-out observations.

For each target and split, the root mean squared prediction error summarizes discrepancies over its observed held-out French months. Each split average gives target families equal weight, and the overall average gives the three split averages equal weight. French-only fitting uses the 159 donors with complete French coverage. The multinational procedure uses smaller, target-specific donor sets satisfying additional

German, US and global coverage requirements, although global traffic does not enter its fitting criterion. The exercise therefore compares the predictive performance of these two procedures, including their donor eligibility rules.[8]

French-only fitting remains the default unless the multinational procedure reduces the average held-out French prediction error by at least two bootstrap standard errors. The standard error of this difference comes from 9,999 paired whole-parent bootstrap draws, resampling the 20 liable parent firms and retaining all website families belonging to each sampled parent. Each draw uses the same parent multiplicities across procedures and validation splits and recomputes the average prediction-error difference from the saved held-out errors. This bootstrap conditions on the fitted predictions; it does not refit the synthetic weights. For the main donor group, the multinational procedure improves prediction error by 0.076 log percentage points, with a bootstrap standard error of 0.482. This improvement falls below the two-standard-error threshold, so French-only fitting remains the default.

### 4.5 Inference for the full estimation pipeline

For the main regression tables and monthly paths, 1,999 pairs-bootstrap draws provide inference. Each draw samples 20 liable parents with replacement, retaining every family and month under a selected parent. Each draw separately samples 469 exposed nonliable families and 169 residual nonliable families with replacement, retaining their full histories and observed missingness. The residual, exposed and pooled donor sets are selected from these sampled family copies using the original French completeness rule.

Each draw recomputes donor-change scales and demeaned pre-announcement histories, fits the nonnegative unit-sum synthetic weights with $\lambda = \sqrt{12}$, realigns the September–November baseline, and reestimates the annual and monthly regressions. Duplicate donor copies are separate coordinates in the simplex and each receives its own ridge penalty, as in the expanded resampled dataset. Available observations and original equal family-month weights are retained. Composition is standardized using the original 55-family mean and SD throughout, preserving the coefficient's unit. Actual-control regressions use the same sampled histories. Standard errors are the sample standard deviation across draws; plotted 95% intervals use the 2.5th and 97.5th percentiles and are pointwise rather than simultaneous.

Model-only profiles use these ordinary parent draws. The observed pure-group raw and EB means instead use the revenue-stratified parent draws described above, repeating the same donor resampling and synthetic fitting. December 2018 is excluded from annual contrasts and primary monthly regressions. In Figures 4–5, it appears only as a separately estimated transition point without an interval. The monthly EB calculation retains its original separate December transition in nuisance fitting, while Figure 6 omits it; the annual noise calibration excludes December.

Appendix H reports an alternative approach that holds fitted synthetic controls fixed when calculating standard errors and confidence intervals.

[8] Donor membership, coverage requirements and advertising-exposure classifications are held fixed across validation splits, rather than reconstructed using only information available at each training date. No held-out traffic values enter the fitting criterion.

## 4.6 Results

**Traffic responses among liable sites** Figure 4 presents monthly French traffic paths and liable-minus-comparison contrasts for the 55 liable website families in the sample. The left column uses individual synthetic controls fitted to prepolicy French traffic only; the right column uses equal-weighted actual controls. Both columns use the comparison group without documented advertising exposure. The upper panels display the liable and comparison paths, and the lower panels display their differences. Table 7 reports the corresponding period DiD coefficients and bootstrap standard errors for actual and synthetic comparisons, using the unexposed control group. Appendix G, Table G.1, reports the corresponding comparisons using advertising-exposed and pooled controls. The table also reports sample counts, fixed effects and regression fit statistics.

The stacked synthetic-control estimate is the preferred specification, using French-only prepolicy fitting and the residual advertising-exposure group. It indicates a traffic decline of 2.81 log percentage points (s.e. 7.78) among liable platforms relative to their synthetic comparisons. Scaling by the 3-percentage-point statutory DST gives a response of $-0.94$ log percentage points of traffic per percentage-point increase in the statutory rate.

Table 7: Difference-in-differences estimates of French traffic responses

| | No documented advertising exposure | |
|---|---|---|
| Model | (1) | (2) |
| Estimator | DiD | Stacked SC |
| Dependent variable | $y_{it}$ | $y_{ist}$ |
| Liable × Post ($\beta$) | +5.74 | −2.81 |
| | (7.56) | (7.78) |
| Liable families | 55 | 55 |
| Liable parent firms | 20 | 20 |
| Raw comparison families | 169 | – |
| Synthetic comparison units | – | 55 |
| Eligible donor pool | – | 159 |
| Eligible donors per liable family | – | 159 |
| Regression units | 224 | 110 |
| Observations | 3,311 | 1,650 |
| Liable / nonliable resampling blocks | 20 / 169 | 20 / 169 |
| Unit fixed effects | Yes | Yes |
| Month fixed effects | Yes | Yes |
| $R^2$ | 0.953 | 0.994 |
| Within $R^2$ | 0.00073 | 0.00101 |
| RMSE (log units) | 0.369 | 0.177 |

*Notes:* January–December 2019 versus September–November 2018; December 2018 excluded. Coefficients are $100\beta$, with standard errors from 1,999 full-history bootstrap draws in parentheses. Column (1) estimates Equation (10) on observed families; column (2) stacks each liable family with its synthetic member, retaining member and shared month fixed effects. Each draw samples 20 whole liable parents and, separately, 169 residual nonliable families with replacement. All family histories and missingness remain together. Eligible donor copies are selected using French coverage; donor-change scales, synthetic weights and baseline offsets are reestimated before refitting the regression.

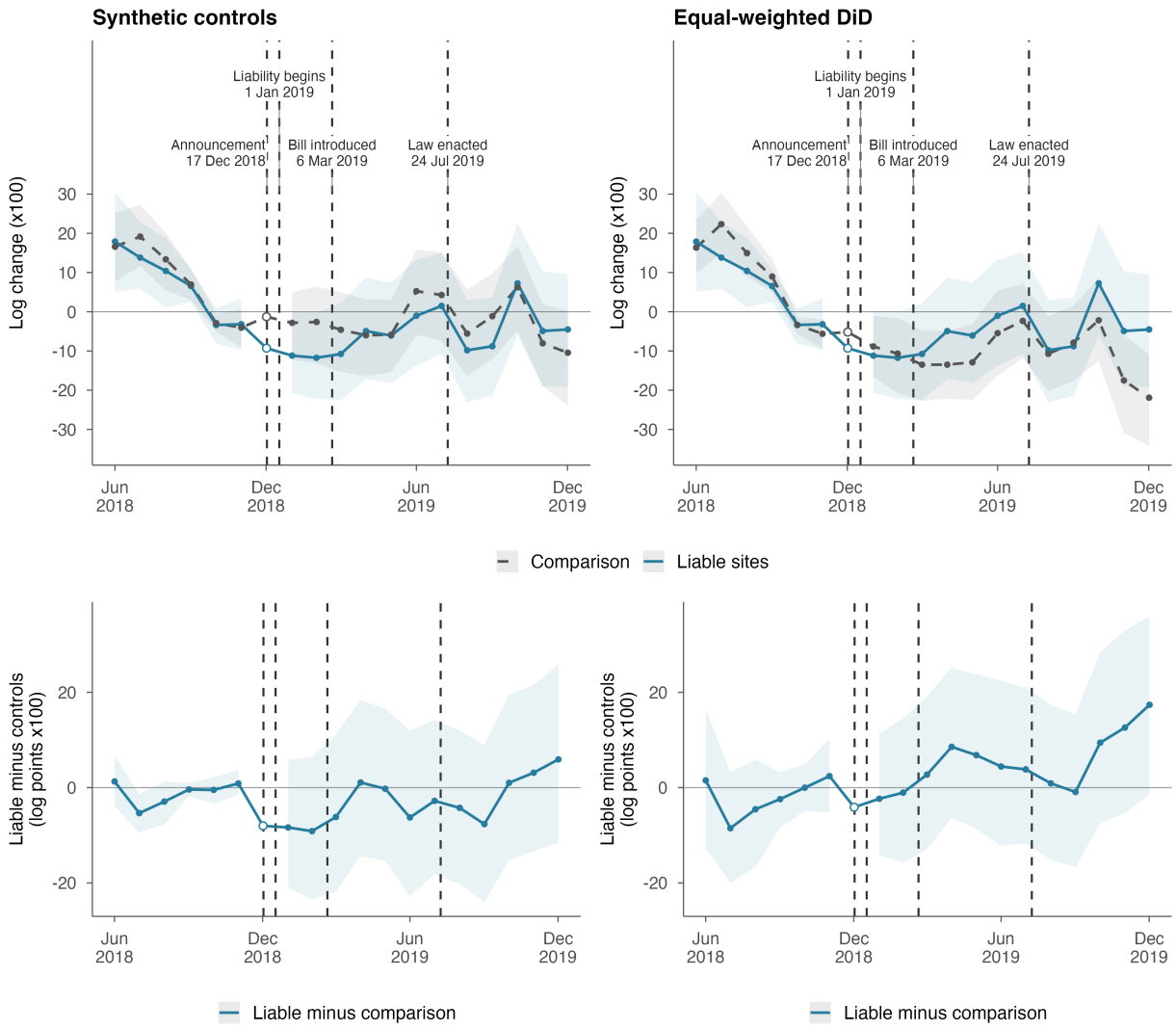



Figure 4: French traffic responses of liable platforms

*Notes:* Left: synthetic controls; right: equal-weighted actual-control DiD. Both use 55 liable families under 20 parents and the unexposed nonliable group (169 families; 159 eligible synthetic donors). Upper panels show liable and comparison paths; lower panels show their differences. Series are centered on September–November 2018 and displayed as 100 times log changes or differences. Shading gives pointwise 95% percentile intervals from 1,999 draws of whole liable parents and residual nonliable family histories. Synthetic scaling, weights, baseline offsets and regressions are reestimated in every draw. December 2018 is display-only and has no interval.

**Traffic responses of advertising-exposed nonliable sites.** Figure 5 compares 469 advertising-exposed nonliable families with unexposed families, using synthetic controls drawn from the unexposed donor pool. Its left column uses individual synthetic controls and its right column uses equal-weighted actual unexposed controls. The upper row displays the exposed and comparison paths; the lower row displays the corresponding exposed-minus-comparison differences. Table 8 reports the exposed-by-post coefficient and bootstrap standard error for equal-weighted actual-control DiD and stacked synthetic DiD, together with sample counts, fixed effects and fit statistics.

The stacked synthetic comparison estimates a traffic decline of 8.24 log percentage points (s.e. 4.96) among advertising-exposed nonliable families. This pattern is consistent with effects transmitted through advertising relationships. Remember, however, that in practice this may include not only reductions in advertising on the liable platforms but also the effect of diminished returns on ads displayed on behalf of liable corporations. The exposure measure can identify websites that buy advertising, display advertising or use associated tracking and measurement technologies; it does not separate the roles or establish which channel generates the traffic decline.

Table 8: Difference-in-differences estimates for advertising-exposed nonliable sites

| Model | (1) | (2) |
|---|---|---|
| Estimator | Conventional DiD | Stacked synthetic DiD |
| Dependent variable | $y_{it}$ | $y_{ist}$ |
| Exposed $\times$ Post ($\delta$) | $-1.75$ | $-8.24$ |
| | (5.30) | (4.96) |
| Exposed nonliable families | 469 | 469 |
| Raw unexposed families | 169 | – |
| Synthetic comparison units | – | 469 |
| Starting eligible donor pool | – | 159 |
| Eligible donors per exposed family | – | 159 |
| Regression units | 638 | 938 |
| Observations | 9,506 | 14,040 |
| Exposed / residual resampling blocks | 469 / 169 | 469 / 169 |
| Unit fixed effects | Yes | Yes |
| Month fixed effects | Yes | Yes |
| $R^2$ | 0.895 | 0.934 |
| Within $R^2$ | 0.00004 | 0.00212 |
| RMSE (log units) | 0.477 | 0.358 |

*Notes:* January–December 2019 versus September–November 2018; December 2018 excluded. Coefficients are $100\delta$, with bootstrap standard errors in parentheses. Column (1) estimates Equation (11) on actual nonliable families; column (2) stacks each exposed family with its own synthetic comparison. In each of 1,999 draws, 469 exposed and 169 residual nonliable families are sampled separately with replacement, keeping every selected family's full history and missingness.

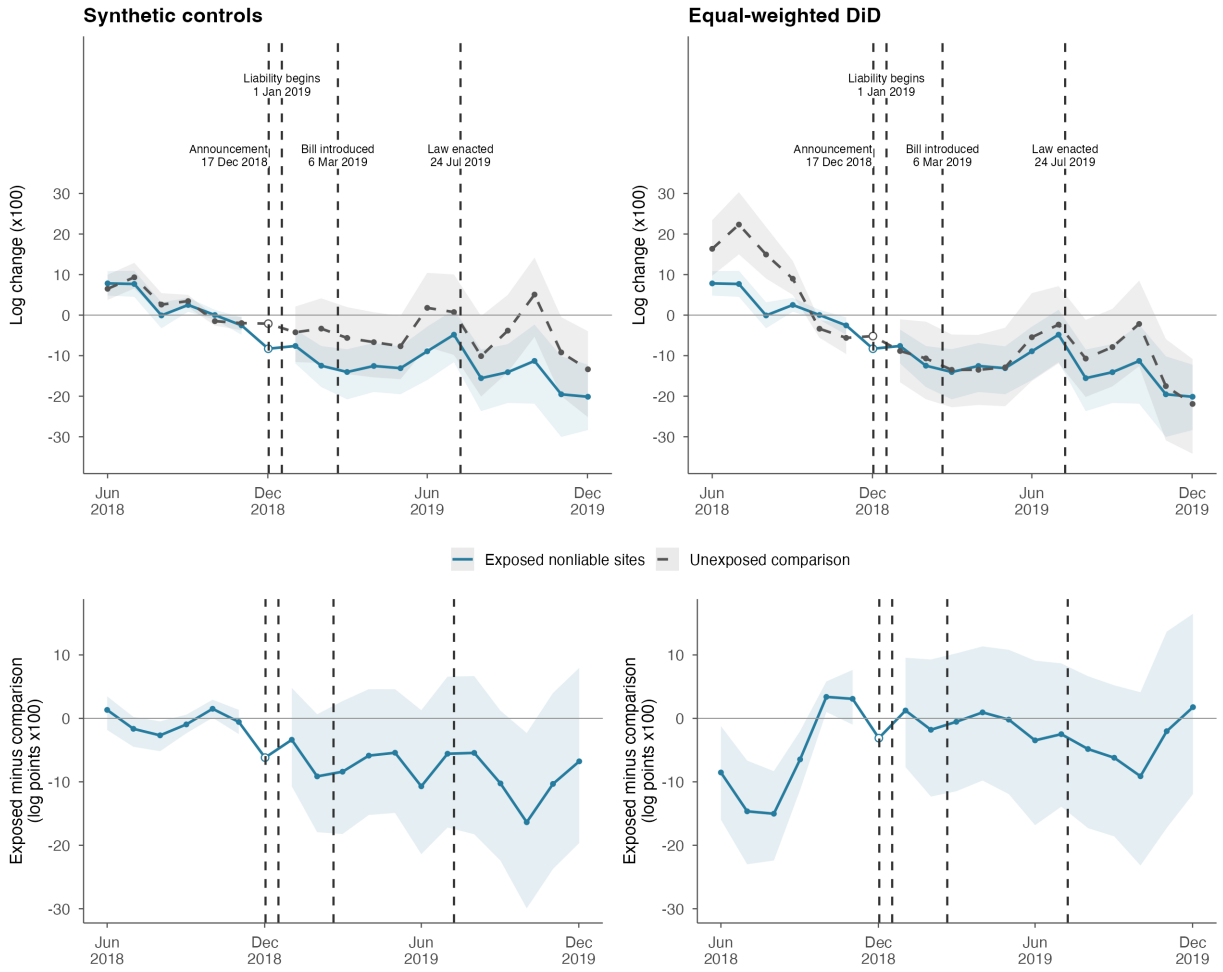



Figure 5: French traffic responses of advertising-exposed nonliable sites

*Notes:* Synthetic controls (left) and equal-weighted actual controls (right) compare 469 exposed nonliable families with 169 residual families; the synthetic pool contains 159 French-complete donors. Upper panels show normalized traffic paths; lower panels show exposed-minus-comparison contrasts. All series are centered on September–November 2018 and measured as 100 times log changes or differences. Shading gives pointwise 95% percentile intervals from 1,999 draws of full family histories within exposure groups. Every draw repeats synthetic scaling, weight estimation, baseline alignment and the regressions, sharing donor multiplicities across target comparisons.

### 4.7 Traffic responses by business model

Table 9 reports $\beta$ and $\theta_C$ from Equation (14), using actual-control and stacked synthetic DiD for the unexposed control group. Appendix Table G.2 reports the same composition specification using advertising-exposed and pooled controls. The table includes bootstrap standard errors, fit statistics, sample counts and fixed-effect indicators. The period regressions compare September–November 2018 with January–December 2019, excluding December 2018. Coefficients are displayed as 100 times log units; the composition gradient is per original liable-sample standard deviation.

Across both DiD and stacked synthetic specifications, a one-standard-deviation increase in advertising's share of in-scope revenue is associated with a 12.38 log percentage point increase in traffic (s.e. 4.92).

To understand this estimate in the context of specific business models, Table 10 reports the corresponding annual pre/post effects from the composition model, raw pure-group averages and empirical Bayes estimates. Appendix E reports the annual raw synthetic-control contrast, composition-model prediction, empirical Bayes estimate, and weight on the raw contrast for each of the 55 liable website families. The hypothetical advertising endpoint is $+16.48$ (s.e. 9.11), and the intermediation endpoint is $-17.50$ (s.e. 10.09). Among observed pure-type families, the raw advertising and intermediation averages are $+22.25$ (s.e. 7.97) and $+5.02$ (s.e. 6.59). The corresponding EB averages are $+16.90$ (s.e. 7.81) and $-11.13$ (s.e. 8.24).

Figure 6 shows the dynamic counterparts of these estimates. They compare monthly composition-model predictions with raw and empirical Bayes averages among observed pure-type website families, as in websites whose parents report only advertising or only intermediation in-scope revenue.

Taken together, these results show that the aggregate estimates presented in Table 7 conceal substantial heterogeneity by parent business model. Moreover, the estimated effects comport with the predictions of the models presented in Section 3. The following section discusses these results more closely.

## 5 Discussion

The empirical results are consistent with the key idea from Section 3, which is that although France's DST framework places advertising and intermediation under the same digital service umbrella, those two services have fundamentally different relationships with user participation that are important to take into account when assessing the impact of DSTs on users. In practice, the empirical results suggest declines in traffic for pure intermediation sites and increases for pure advertising sites. Moreover, omitting this source of heterogeneity would have provided only an incomplete understanding of the heterogeneous effects of the DST on web traffic and user participation.

Seen through the lens of the models, these traffic responses can be mapped to unobservable platform decisions.

Consider first advertising-funded platforms. The empirical Bayes average among observed pure ad-

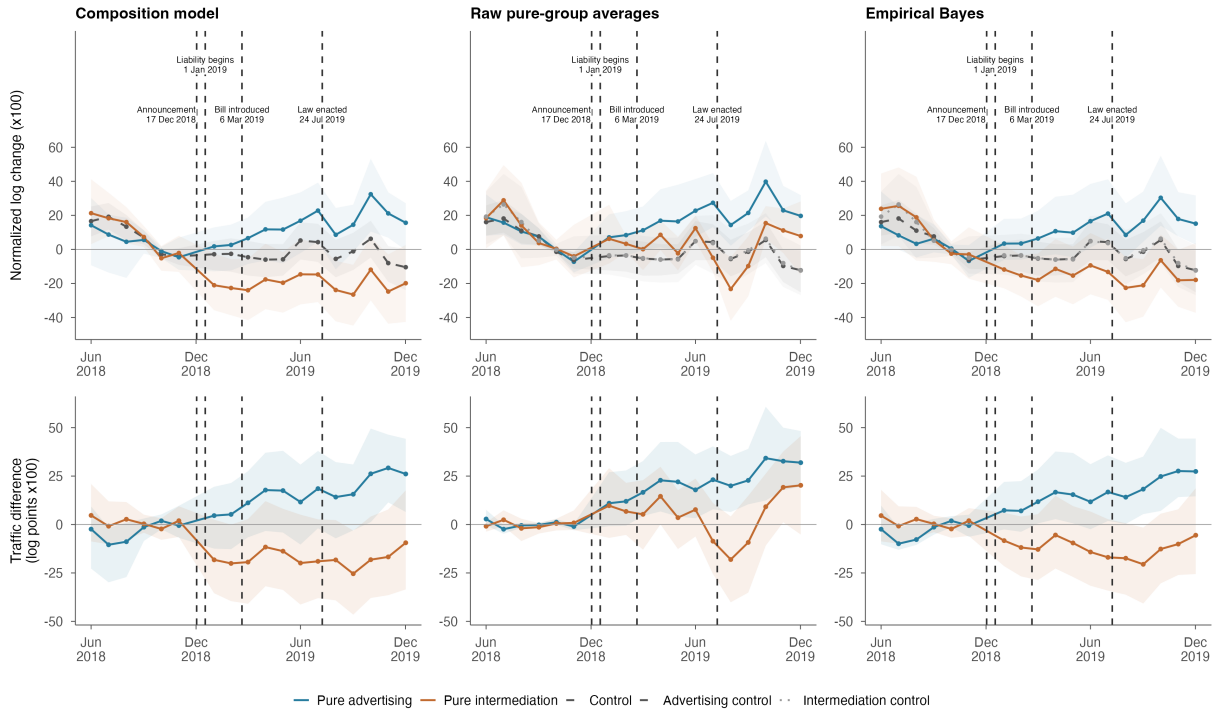



Figure 6: Advertising composition and traffic

*Notes:* Left: hypothetical pure advertising ($C = 1$, blue) and intermediation ($C = 0$, orange) profiles from the original stacked composition model (55 families, 20 parents). Middle: raw means among 13 observed pure advertising and 5 pure intermediation families. Right: EB means for the same groups, centered on the original model. Upper panels show normalized traffic paths and synthetic comparisons; lower panels show gaps. Paths are centered on September–November 2018 in 100 times log units; December is omitted. Shading gives pointwise 95% percentile intervals from 1,999 draws, with synthetic scaling, weights, baseline offsets and model coefficients refitted. The model uses ordinary whole-parent draws; raw and EB columns use revenue-stratified parent draws. Each draw samples residual nonliable donor families with shared multiplicities across targets. EB additionally refits noise correlation, website scales, monthly signal variances and reliability weights before recentering.

Table 9: Traffic responses by advertising composition of in-scope revenue

| | No documented advertising exposure | |
|---|---|---|
| Estimator | Actual DiD | Stacked SC |
| Model | (1) | (2) |
| Liable × Post ($\beta$) | $+5.74$ | $-2.81$ |
| | (6.72) | (6.99) |
| Liable × Post × Composition ($\theta_C$) | $+12.38$ | $+12.38$ |
| | (4.92) | (4.92) |
| $R^2$ | 0.953 | 0.994 |
| Within $R^2$ | 0.00513 | 0.03948 |
| RMSE (log units) | 0.368 | 0.173 |
| Liable families / parents | 55 / 20 | 55 / 20 |
| Raw comparison families | 169 | – |
| Synthetic comparison units | – | 55 |
| Observations | 3,311 | 1,650 |
| Regression units | 224 | 110 |
| Liable / nonliable resampling blocks | 20 / 169 | 20 / 169 |
| Unit fixed effects | Yes | Yes |
| Month fixed effects | Yes | Yes |

*Notes:* Equation (14); September–November 2018 versus January–December 2019, excluding December 2018. Coefficients and standard errors are multiplied by 100. Composition retains its original 55-family mean and SD, so $\beta$ is the change at original mean composition and $\theta_C$ is the gradient per original-sample SD. Standard errors use 1,999 draws of 20 whole liable parents and 169 residual nonliable families. Every draw reestimates synthetic scaling, weights, baseline offsets and both regression coefficients; actual-control columns use the same resampled histories. The synthetic comparison changes the common intercept, while the gradient is determined by liable outcomes and is unchanged by refitting weights within a draw. Counts and fit statistics describe the original sample.

Table 10: Pre/post traffic effects by business model and estimation method

| | Composition model | Raw group average | Empirical Bayes |
|---|---|---|---|
| Pure advertising | $+16.48$ | $+22.25$ | $+16.90$ |
| | (9.11) | (7.97) | (7.81) |
| Pure intermediation | $-17.50$ | $+5.02$ | $-11.13$ |
| | (10.09) | (6.59) | (8.24) |

*Notes:* Annual effects in log percentage points, with bootstrap standard errors in parentheses. September–November 2018 versus January–December 2019; December 2018 excluded. The composition model evaluates the original stacked specification at $C = 1$ and $C = 0$, using 55 families under 20 parents. Raw and EB columns average equally over 13 observed pure advertising families or 5 pure intermediation families; EB centers use the original composition model. Each column uses 1,999 draws with synthetic donor weights refitted. The model column uses ordinary whole-parent draws; raw and EB columns use 5 advertising, 10 mixed and 5 intermediation parent draws within fixed revenue groups. All draws independently sample the 169 residual nonliable families, retaining complete histories. Scaling, weights, baseline offsets and model coefficients are reestimated; EB additionally refits AR(1) correlation, website noise scales, residual signal variance and reliability weights. Annual EB is calculated directly from annual contrasts.

vertising websites is an increase of 16.9 log percentage points (s.e. 7.8). Under the maintained model assumptions, a positive traffic effect requires the benefit of reduced advertising intensity to dominate the loss from lower content and service provision in Equation (8).

The advertising and content decomposition can be developed further using the estimated 8.24 log percentage point traffic decline at advertising-exposed nonliable websites.[9] Combining this quantity proxy with Acun et al. (2026)'s estimated 3.4 percent increase in effective costs per click, approximated as 0.034 log points, gives an illustrative advertising-demand elasticity:

$$\varepsilon_{A,p} \simeq \frac{\Delta \log A}{\Delta \log p} = \frac{-0.0824}{0.034} \simeq -2.42. \tag{24}$$

It is also possible to leverage the model to decompose the change in traffic to advertising platforms into the contribution from changing advertising intensity and changes in content and service provision. Again assuming that the change in traffic to third party sites as an estimate of the change in total advertising impressions, since $a = A/L$:

$$\Delta \log a = \Delta \log A - \Delta \log L = -0.0824 - 0.1690 = -0.2514. \tag{25}$$

Log linearizing Equation (8), and using the unit elasticity of traffic with respect to advertising intensity implied by the expenditure-minimization condition in Appendix C.2, yields an expression for estimating the content effect:

$$\Delta \log L \approx -\Delta \log a + \underbrace{\varepsilon_{L,g_U} \Delta \log g_U}_{\text{content/services effect}}, \tag{26}$$

Lower advertising intensity therefore contributes approximately $+25.14$ log percentage points to advertising platform traffic, while its net traffic effect is $+16.90$. The implied residual contribution from content and services is $-8.24$ log percentage points.

The intermediation point estimates are consistent with the model's negative traffic prediction. Langenmayr and Muddasani (2025) report a French consumer log-price effect of 0.022, or 2.2 log percentage points, for products sold by third-party sellers using Fulfillment by Amazon relative to identical products sold directly by Amazon. Combining that price response with Amazon's empirical Bayes traffic decline of 4.50 log percentage points gives an illustrative upper bound on the magnitude of the traffic-price elasticity when both higher prices and reduced seller availability weakly reduce traffic:

$$0 \leq |\varepsilon_{L,p_B}| \leq \frac{-\Delta \log L}{\Delta \log p_B} \simeq \frac{0.0450}{0.022} = 2.05. \tag{27}$$

This estimate is a touch larger than prominent e-commerce price estimates of demand in the Economics

[9] The following requires assuming that the decline in traffic to third party websites is a proportional measure of the change in advertising impressions, holding click-through rates and the conversion of clicks into measured visits fixed. Translating a change in cost per click into the model's price per impression also requires a stable click-through rate.

and Finance literature, where Houde et al. (2023) estimates e-commerce price elasticities of -1.52, Einav et al. (2014) estimates -1.67, and Baugh et al. (2018) estimates -1.2 to -1.4. Assuming that demand elasticities appear proportionately in traffic, these magnitudes imply that approximately 18%–41% of the estimated traffic response reflects declines in product availability or other declines in platform quality. The exercise illustrates how traffic can complement price evidence while making clear the additional assumptions needed to distinguish the channels.

Extending this calculation to pure intermediation websites, suppose that the buyer-price increase of 2.2 log percentage points estimated by Langenmayr and Muddasani (2025) applies uniformly across these websites. Combining this price response with their empirical Bayes average traffic decline of 11.13 log percentage points (s.e. 8.24) yields an illustrative upper bound of 5.06 on the absolute traffic-price elasticity. Then, ssuming that demand elasticities translate proportionately into traffic responses, the benchmark absolute elasticities of 1.2–1.67 discussed above imply that higher prices account for approximately 24%–33% of the estimated traffic decline. The remaining 67%–76% is attributable, within the model, to reduced seller availability or other declines in platform quality.

# 6 Conclusion

Digital services taxes change the return to attracting and serving users, with consequences that depend on how platforms earn revenue. This paper studies those consequences around the December 2018 announcement of France's DST, using web traffic to compare user participation across platforms with different business models. Relative to synthetic controls constructed from nonliable websites without observed connections to liable-owned advertising technologies, the estimates indicate an average traffic decline of 2.8 log percentage points (s.e. 7.78) across the 55 liable website families owned by 20 corporate groups.

The traffic response varies with the platform's business model. A one-standard-deviation increase in advertising's share of in-scope revenue is associated with a 12.38 log percentage point increase in traffic (s.e. 4.92). Empirical Bayes estimates indicate an increase of 16.9 log percentage points (s.e. 7.8) for pure advertising websites and a decline of 11.1 log percentage points (s.e. 8.2) for pure intermediation websites. The theoretical framework connects these contrasting responses to platform incentives. For transaction intermediaries, higher buyer prices and reduced seller supply discourage visits. For advertising-funded platforms, reduced advertising intensity can attract users even as lower expenditure on content and services discourages participation.

Combining the traffic estimates with existing price evidence illustrates how this approach can shed light on platform choices. The decline in traffic to advertising-exposed nonliable websites and the external advertising-price response yield an illustrative advertising-demand elasticity of approximately $-2.4$. Together with the advertising platform empirical Bayes traffic increase of 16.9 log percentage points, the calculation implies a decline in advertising intensity of approximately 25.14 log percentage points. Within the model, reduced advertising contributes approximately $+25.14$ log percentage points to traffic, offset by a content-and-services contribution of $-8.24$ log percentage points. Amazon's empirical Bayes traffic decline of 4.50 log percentage points, combined with the external consumer-price response, gives a traffic-price elasticity of approximately $-2.05$ if the entire decline is attributed to prices, or an upper bound of 2.05 on its absolute magnitude when reduced seller availability also discourages visits. Extrapolating to intermediation websites as a whole, the equivalent elasticity is -5.5. Finally, assuming that demand elasticities from the digital economics literature appear proportionately in traffic, the effects suggest that 18%-24% of Amazon traffic and 67%-76% of overall intermediation traffic are attributable to reduced seller availability or other declines in local platform quality.

The findings bear on proposals to tax advertising as a means of mitigating the harms of social media (Romer, 2019; Acemoglu and Johnson, 2024). By reducing advertising intensity, such taxes can make platforms more attractive and increase participation, even as they discourage investment in user services. Advertising taxes may therefore be a counterproductive substitute for levies that directly target harmful use, consistent with the targeting principle (Sandmo, 1975; Kopczuk, 2003). This possibility reflects the logic of the second best: when harmful use cannot be targeted directly, reducing advertising need not improve welfare if it also exacerbates overconsumption of digital services (Lipsey and Lancaster, 1956). Assessing

these proposals consequently requires weighing any corrective benefits in the advertising market against their effects on platform use and service provision. This is a promising direction for future normative work.

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